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SUMMER INTERNSHIP PROJECT REPORT

Credit Risk Segmentation & NPA Reduction Strategy for GrayQuest Education Finance

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A note on the data: all analysis is based on internship datasets used for academic purposes. The findings and recommendations are entirely my own and do not represent the official position of GrayQuest Education Finance Pvt. Ltd.

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CERTIFICATE OF COMPLETION

It is certified that Mr Liash Roy, a PGDM student of Great Lakes Institute of Management, Gurgaon has undergone a Summer Internship at GrayQuest Education Finance Pvt. Ltd. From 6th April ,2025 to 5th June , 2025.

His project titled “***Credit Risk Segmentation & NPA Reduction Strategy for GrayQuest Education Finance***” has been completed to our satisfaction and the present report is a bona fide account of the same.



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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AIC	Akaike Information Criterion
AP	Andhra Pradesh
AUC	Area Under Curve
BFGS	Broyden–Fletcher–Goldfarb–Shanno (optimisation algorithm)
CIBIL	Credit Information Bureau (India) Limited
CR	Credit Review
DID	Difference-in-Differences
DPD	Days Past Due
EDA	Exploratory Data Analysis
EMI	Equated Monthly Instalment
EPFO	Employees' Provident Fund Organisation
EWS	Early Warning System
FOIR	Fixed Obligations to Income Ratio
FPR	False Positive Rate
FY	Financial Year
GLIM	Great Lakes Institute of Management
GST	Goods and Services Tax
H0	Null Hypothesis
H1	Alternative Hypothesis
HSA	Highest Sanctioned Amount
HQ	Headquarters
IRAC	Income Recognition, Asset Classification and Provisioning
ITR	Income Tax Return
KPI	Key Performance Indicator
LE	Loan Execution Stage
MH	Maharashtra
MLE	Maximum Likelihood Estimation
MoM	Month-on-Month
NACH	National Automated Clearing House
NBFC	Non-Banking Financial Company

Abbreviation	Full Form
NPA	Non-Performing Asset
NTC	New To Credit
OR	Odds Ratio
PGDM	Post Graduate Diploma in Management
pp	Percentage Points
RBI	Reserve Bank of India
ROC	Receiver Operating Characteristic
ROC-AUC	Receiver Operating Characteristic — Area Under Curve
ROE	Return on Equity
SBR	Scale-Based Regulation
SC	Sri Chaitanya Educational Institutions
SE	Self-Employed
SMA	Special Mention Account
SOP	Standard Operating Procedure
SIP	Summer Internship Project
TG	Telangana
TN	True Negatives
TP	True Positives
TPR	True Positive Rate
VH	Very High Risk
VIF	Variance Inflation Factor

EXECUTIVE SUMMARY

GrayQuest Education Finance's portfolio NPA rate of 1.34% looks acceptable on the surface. However, a closer look at the underlying segments shows otherwise. Inside that number, the manual approval channel defaults at 2.36% while the bureau channel defaults at 0.59%. The 11-EMI product defaults at nearly six times the rate of the 6-EMI product. GQ_Defaulter profiles — customers with prior GrayQuest default history — were still being approved through manual overrides and defaulting at 7.36%. These patterns are not isolated cases. They reflect consistent and identifiable patterns that can potentially be addressed.

This report analyses 24,626 loan application records and 181 NPA customers across two institutional partners — Narayana Educational Institutions and Sri Chaitanya Educational Institutions — using binary logistic regression (Python, statsmodels, AUC = 0.787), hypothesis testing, and a tiered approval simulation across six policy scenarios. The analytical outputs are a four-dimension Credit Risk Scorecard, a rule-based Early Warning System, and a set of specific policy recommendations with quantified impact estimates.

Top 3 Findings

#	Finding	Key Statistic
1	The manual approval channel emerges as one of the most significant controllable drivers of NPAs. It generates 4x the default rate of bureau approvals (2.36% vs 0.59%) and accounts for 74.7% of all NPAs despite covering only 42.5% of disbursements. This results in approximately 97 excess defaults and ₹55.4 lakh in avoidable NPA outstanding per cycle — driven by a process that actively overrides the bureau algorithm's risk signals.	OR_manual = 2.21 (p<0.001) 74.7% of all NPAs ₹55.4L avoidable
2	Tenure appears as the strongest predictor of default in the regression model, indicating a potential product design concern in addition to borrower-level risk. The 11-EMI product defaults at 2.45% versus 0.43% for 6-EMI (5.7x higher). It accounts for 54.6% of all observed defaults while covering 29.9% of disbursements. Restricting 11-EMI access does not require better underwriting — it requires a product policy decision.	OR_Tenure11 = 3.97 (p<0.001, z=6.14) 11-EMI = 5.7x 6-EMI rate
3	55.2% of all NPA accounts — 100 of 181 — sit in the 30–60 DPD band (SMA-2 under RBI norms) and have not yet crystallised into confirmed NPAs. Average outstanding in this band is ₹59,954. Sri Chaitanya has 29 accounts in 90+ DPD versus Narayana's 7 — a direct consequence of SC's higher mid-month NACH failure rate (11.55% vs 9.87%). This is a collection execution gap, not a credit quality gap.	₹59.95L outstanding in 30–60 DPD SC: 29 cases in 90+ DPD vs Narayana: 7

Executive Summary — Top 3 Findings

Top 3 Recommendations

#	Action	Expected Impact	Timeline
1	Deploy the 4-dimension Credit Risk Scorecard as the mandatory pre-approval gate. Auto-reject all Score ≥ 6 applications (5.9% of portfolio) with no manual override permitted. GQ_Defaulters profiles: permanent block until all prior dues are cleared. Apply Conditional Approval with co-applicant and income proof for Score 2–3 (14.6% of portfolio).	NPA rate: 1.34% $\rightarrow$ $\sim$ 0.91% (Scenario F) 57 defaults prevented per cycle ₹32.5L NPA saved	Month 1–4
2	Activate same-day NACH failure response: automated contact within 24 hours of the first NACH failure; field agent assigned on the second consecutive failure; manager escalation for any outstanding above ₹50,000. Sri Chaitanya's mid-month collection cycle (11.55% failure rate) gets a dedicated collection resource from Month 1.	Prevents 30–60 DPD escalation Targets ₹59.95L in recoverable SMA-2 outstanding	Month 1–2
3	Restrict 11-EMI to salaried borrowers with CIBIL ≥ 750 , effective immediately. Suspend new disbursements through Ratnaafin Lite pending underwriting review — it defaults at 5.15%, the highest of any NBFC partner, and generates 23% of all defaults from 6% of disbursements. Route all Low Risk (Score 0–1) applications through Mirae Asset (0.51% default rate, largest partner by volume).	Structural NPA reduction through product and partner policy Ratnaafin Lite: 23% of defaults from 6% of loans	Month 2–6

Executive Summary — Top 3 Recommendations

The Scorecard, logistic regression model, and Early Warning System are ready for deployment. These recommendations largely involve process-level changes in credit and collection workflows rather than requiring new technical infrastructure. The NPA issue does not appear to be uniform across the entire portfolio. Instead, it is concentrated within a relatively small and identifiable segment of the customer base ($\sim$ 9–10%). This report demonstrates that such segments can be identified with reasonable clarity. The key consideration is how effectively approval and collection processes are structured to act on these insights.

CHAPTER 1 — INTRODUCTION

1.1 About GrayQuest Education Finance

GrayQuest is a Mumbai-based fintech servicer — not an NBFC itself — that originates, underwrites, and processes fee EMI loans on behalf of regulated NBFC partners. Founded in 2017, it has built a platform covering ~5,000 institutions and ~2 million borrowers. Revenue of ~₹24.6 crore in FY2025 (37% YoY growth) reflects rapid scale-up, funded by ~\$27.6 million raised across five rounds including a 2024 Series B (Tracxn, 2025).

Parameter	Detail
Founded	March 2017 Mahalaxmi, Mumbai, Maharashtra
Founders	Rishab Mehta (CEO) & Rahul Mehta (Co-Founder)
Business Classification	Fintech servicer — not itself an NBFC; capital deployed through partner NBFCs
Reported Funding	~\$27.6 million across 5 rounds (Seed → Series B-II, 2020–2024)
Key Investors	Pravega Ventures, IIFL Fintech Fund, Claypond Capital, Mirae Asset
Institutional Partners	~5,000+ schools, colleges & universities (as reported)
Reported FY2025 Revenue	~₹24.6 crore (~37% YoY growth, Tracxn, 2025)
Reported Team Size	~250 employees (publicly available, 2025)

Table 1.1 — GrayQuest Education Finance: Company Snapshot (Sources: publicly available)

Business Model

GrayQuest deploys NBFC partners' capital, not its own. It manages everything else: institutional onboarding, credit underwriting, NBFC assignment, and EMI collection via NACH. The core product is a 0-cost EMI plan — 6, 9, or 11 instalments, no interest charged to the borrower. Revenue comes from service fees paid by institutional and lending partners.

Product / Feature	Description
Fee EMI Plan (Core)	0-cost EMI across 6, 9, or 11 months — no interest charged to borrower
ARC Platform (SaaS)	Institutional fee management, communication & reconciliation software
GQ+ Rewards	Loyalty benefits programme for EMI-paying parents
Insurance Add-ons	Curated insurance products bundled with fee financing (eligible applicants)

Table 1.2 — Core Product Offerings

NBFC & Lending Partners

All disbursements flow through regulated NBFC partners. Default rates below are from the internship dataset — indicative, not audited.

NBFC Partner	Segment Served	Observed Default Rate	Risk Signal
Mirae Asset Financial Services	Salaried / bureau-approved profiles	~0.51%	Low
Fintree Finance	Specific institutional partnerships	~0.65%	Low
Arka Fincap (Kirloskar Group)	Verified income profiles	~1.19%	Low–Moderate
Western Capital	Semi-urban borrower segments	~1.59%	Moderate
Ratnaafin Capital (Standard)	Moderate-risk applicant profiles	~1.96%	Moderate
Ratnaafin Capital (Lite product)	Higher-risk / relaxed underwriting criteria	~5.15%	High

Table 1.3 — NBFC Partner Observed Default Rates (Internship Dataset) | Green = Low • Yellow = Moderate
• Red = High

Ratnaafin Lite stands out at 5.15% — 3.8x the portfolio average. Mirae Asset at 0.51% is the safest. These differences are explored in Chapter 4.

1.2 Model of Operations

GrayQuest's loan lifecycle runs across eight stages — from initial lead capture through to loan closure or NPA management. The workflow below is based on internship observations.

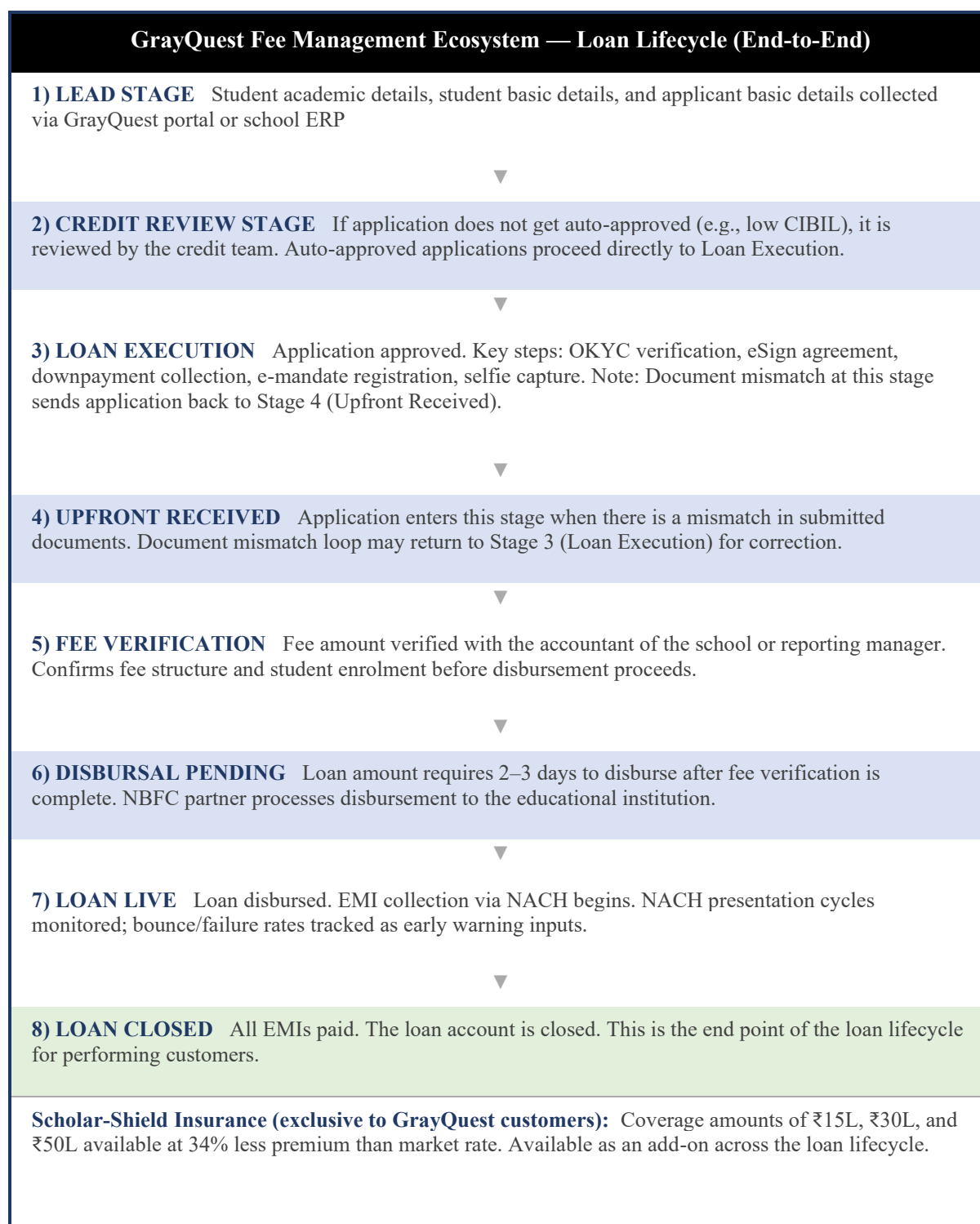


Figure 1.1 — GrayQuest Fee Management Ecosystem: Official 8-Stage Loan Lifecycle (Source: GrayQuest company documentation)

1.3 Financial Performance Overview

Key financial and portfolio indicators — from public sources and internship data, not audited figures.

Metric	Indicative Value	Basis
FY2025 Revenue	~₹24.6 Crore	Publicly reported; ~37% YoY (Tracxn, 2025)
Total Funding	~\$27.6 million	5 rounds: Seed–Series B-II (2020–2024), public sources
Institutional Partners	~5,000+	Company website / LinkedIn
Students / Parents Served	~2 million	Company website (estimated)
Dataset — Applications	~24,626 records	Internship dataset (Narayana + Sri Chaitanya)
Dataset — Disbursed Loans	~13,011 records	Internship dataset
Observed NPA Rate (Dataset)	~1.34%	174 unique defaults / 13,011 disbursed loans
Industry NPA Benchmark (est.)	~2–5%	RBI NBFC sector reports; broad contextual estimate only

Table 1.5 — Financial & Portfolio Indicators (Publicly available + internship data)

The 1.34% overall NPA rate is comparatively low versus the industry benchmark of 2–5%. But the average masks wide internal variation — by NBFC partner, income type, tenure, and approval channel. That variation is what this project analyses.

1.4 Title of the Project

Credit Risk Segmentation & NPA Reduction Strategy for GrayQuest Education Finance

1.5 Need of the Study

GrayQuest's manual approval channel defaults at 4x the bureau rate and drives 74.7% of all NPAs. Nine percent of the portfolio accounts for 43% of defaults. There is no structured scorecard to filter these risks before disbursement. The three data patterns below spell out the problem:

Observation	Dataset Finding	Implication for Study
Manual Approval Risk	Manual channel: ~2.36% default rate vs ~0.59% bureau-based; ~75% of NPAs traced to manual approvals	Suggests override of data-driven decisions may be a contributor to NPA outcomes
No Structured Scorecard	No documented scoring framework observed; outcomes vary by income type, tenure, and institution	Absence of a formal model may contribute to inconsistent underwriting quality
Risk Concentration	~9% of portfolio (High/Very High risk bands) associated with ~43% of observed defaults	Identifiable, concentrated risk segments may be addressable through tiered approval policies

Table 1.6 — Structural Risk Observations Motivating This Study

Objective: Score every applicant before disbursement. Flag at-risk accounts before they default. Simulate the impact. Reduce NPAs from 1.34% to under 1%.

1.6 Objectives of the Study

#	Objective
1	Calculate and benchmark GrayQuest's overall NPA rate across both institutional partners (Narayana and Sri Chaitanya) against available industry estimates for education-sector NBFCs.
2	Identify and quantify key drivers of default — including CIBIL score, income type, tenure, approval channel, and NBFC partner — in the dataset.
3	Develop a multi-dimensional Credit Risk Scorecard classifying applicants into Low, Medium, High, and Very High risk tiers using observable, replicable scoring criteria.
4	Validate the statistical significance of identified risk drivers using Welch's t-test, Z-test, one-way ANOVA, and chi-square tests (significance level $\alpha = 0.05$).
5	Build a logistic regression model estimating individual default probability, evaluated using ROC-AUC and McFadden's Pseudo R^2 .
6	Design a rule-based Early Warning System (EWS) using bureau flags and NACH collection data to enable identification of at-risk accounts before NPA crystallisation.
7	Simulate a tiered approval strategy based on the scorecard and estimate its potential impact on portfolio NPA rate and recoverable outstanding — acknowledging dataset scope limitations.

Table 1.7 — Study Objectives

1.7 Methodology

All analyses run in Python (statsmodels, sklearn, pandas) and cross-validated in Excel. Three datasets were used:

Data Sources

Dataset	Records	Key Variables / Notes
Master_Raw_Data	~24,626 applications	~48 variables: CIBIL score, income type, tenure, approval type, NBFC, disbursement status, etc.
NPA_Master_Data	181 rows (174 unique customers)	DPD, outstanding, repayment status — 7 customers appear at both institutions (NPA recorded twice)
NACH_Summary	6 records	Presentation cycle data — NACH success and failure rates by institution and cycle date

Table 1.8 — Internship Dataset Summary

Default_Flag (1 = NPA, 0 = performing) was built by matching customer IDs across the two datasets. 174 unique defaults are used for rate calculations; 181 appears in the NPA file because 7 customers hold loans at both institutions and are counted twice.

Analytical Methods

Analysis Type	Method / Test	Tool
Data Cleaning & Engineering	Standardisation, log transforms, binary dummies, z-scores	Python — pandas, numpy
Descriptive Analysis	NPA rates, cross-tabulations, distribution profiles	Python + Excel
Hypothesis Testing	Welch's t-test, Z-test, one-way ANOVA, chi-square, Mann-Whitney U	Python — scipy.stats
Correlation Analysis	Point-Biserial and Spearman rank correlation with Default_Flag	Python — scipy.stats
Predictive Modelling	Binary Logistic Regression — MLE via BFGS (67 iterations, $n \approx 12,784$)	Python — statsmodels.api.Logit
Model Evaluation	ROC-AUC, McFadden Pseudo R^2 , AIC, Confusion Matrix (threshold = 0.055)	Python — sklearn.metrics
Risk Scorecard	4-dimension additive scoring model calibrated on observed default rates	Excel (cross-validated in Python)
Approval Simulation	Tiered approval scenario modelling — Scenarios A through F	Excel

Table 1.9 — Analytical Methods & Tools

The logistic regression was fitted on 12,784 records (227 excluded for missing CIBIL). Classification threshold set at 0.055 via F1-score maximisation — the default 0.50 threshold fails at a 1.31% event rate. No train-test split was applied. The model is a policy simulation tool, not a production predictor.

1.8 Scope & Limitations

Scope

- Dataset covers Narayana Educational Institutions and Sri Chaitanya Educational Institutions, primarily across Telangana, Andhra Pradesh, and Maharashtra.
- Analysis spans the full application funnel; NPA analysis is focused on the 13,011 disbursed loan subset.
- Scorecard and tiered approval recommendations are framed as operational inputs for consideration, subject to validation on a broader portfolio.

Limitations

Limitation	Description	Impact on Findings
Dataset Scope	Two institutions; ~13,011 disbursed records	Findings may not generalise to GrayQuest's full ~5,000+ partner portfolio
Missing CIBIL Data	~227 records excluded from regression (missing CIBIL)	Regression sample reduced to ~12,784; NTC segment partially underrepresented
Single NACH Cycle	NACH data covers one presentation cycle only	Longitudinal collection behaviour cannot be assessed
No Macro Variables	Regional income, interest rate, and household debt data not included	Model may not capture macro-driven default clusters
No Train-Test Split	Full disbursed dataset used for model fitting due to low event rate	Out-of-sample predictive accuracy cannot be formally assessed
Cross-Sectional Design	Observational data; associations identified are not causal	Policy implications are directional; causal claims require experimental or panel design

Table 1.10 — Summary of Study Limitations

CHAPTER 2 — LITERATURE REVIEW

2.1 Credit Risk Modelling in NBFCs

NBFCs face structurally higher credit risk than commercial banks because they serve segments that formal banking largely bypasses — self-employed borrowers, thin-bureau profiles, and new-to-credit customers (Behr & Güttler, 2007). The evidence from Indian NBFC studies is consistent: institutions with data-driven underwriting tend to outperform those that rely on manual judgement (Bansal, 2024; RBI, 2023). On technique selection, ensemble models like Random Forest can achieve AUC scores of ~ 0.95 in controlled settings (Bansal, 2024), but logistic regression has held its ground on interpretability — which matters in regulated environments where credit decisions must be explainable to regulators and applicants (Siddiqi, 2017; Thomas, Edelman, & Crook, 2002). Das and Uppal (2021) further found that added model complexity did not reliably improve performance on well-structured Indian credit data.

This study uses logistic regression as the primary model — not because it is definitively superior, but because interpretability and regulatory defensibility matter more here than raw accuracy. The model feeds directly into the scorecard and tiered approval framework in Chapters 4 and 7, and is designed as a policy simulation tool, not a production predictor.

2.2 Education Loan Default Determinants: Indian Context

Indian education loan research has consistently pointed to income stability as a primary default driver. Self-employed borrowers default at higher rates than salaried ones — attributed to income volatility and the difficulty of verifying repayment capacity (Kumar, Ravi, & Anand, 2018; Bhaskaran, Srinivasan, & Thomas, 2016). Bandyopadhyay (2016) notes that SE borrowers show qualitatively different repayment behaviour, not just higher default frequency. In this dataset, the SE default rate was 2.69% versus 0.95% for salaried borrowers — a gap significant at $p < 0.001$ (Z-test) and one of the four dimensions built into the scorecard.

On tenure, Srinivasan and Misra (2019) find that longer EMI cycles widen the window for income shocks. Kumar et al. (2018) add that absolute loan amount is a weak predictor once repayment capacity is controlled — it is the income-to-EMI ratio that matters. The pattern here was identical: the 11-EMI product defaulted at 2.45% versus 0.43% for 6-EMI (Odds Ratio ~ 3.83 in the regression), while default rates across loan amount buckets ranged only from 1.07% to 1.59% with no clear direction.

2.3 CIBIL Score as a Predictor of Default

The CIBIL TransUnion Score (300–900) is the standard bureau instrument used by Indian lenders. Scores above 750 are widely treated as low-risk; below 700 is where default probability rises meaningfully (TransUnion CIBIL, 2022; RBI, 2021). That negative relationship holds in this dataset too — CIBIL < 650 defaulted at ~6.25%, while > 750 came in at ~0.73%. But Acharya & Richardson (2009), Bandyopadhyay (2016), and Kapoor & Goel (2017) all caution against using CIBIL as a standalone gate, and this study found exactly why: a self-employed borrower in the 650–700 band defaulted at 7.14%, while a salaried borrower above 750 came in at 0.56%. The two-point spread between those profiles is not visible in a CIBIL score alone, which is precisely the case for a multi-dimensional scorecard.

New-to-Credit borrowers — no bureau history — are a separate challenge. Kapoor & Goel (2017) treat NTC as broadly equivalent to the 700–750 band in risk. The NTC default rate here was 2.64%, consistent with that framing and reflected in their placement as a moderate-risk (2-point) scorecard entry. NTC borrowers are partially underrepresented in the regression sample because their missing CIBIL scores triggered exclusion.

2.4 Logistic Regression in Credit Scoring

Logistic regression has been applied in credit scoring since Altman (1968) and remains in widespread use due to its interpretability, regulatory acceptance, and competitive performance on structured binary outcome data (Hand & Henley, 1997; Thomas et al., 2002). It estimates default probability via Maximum Likelihood Estimation (MLE), producing coefficients interpretable as log-odds of default for each predictor.

A recent comparative study (Manyumbu, Mushore, & Maposa, 2024) found that logistic regression achieved competitive F1 score, accuracy, and AUC relative to machine learning alternatives on credit registry data. Das and Uppal (2021) similarly found it performed well on Indian NPA prediction tasks and noted that model complexity did not consistently improve performance on well-structured datasets. Model fit is assessed using ROC-AUC (acceptable: ≥ 0.75 ; Hosmer & Lemeshow, 2013) and McFadden Pseudo R^2 (meaningful: > 0.10 ; McFadden, 1974).

Metric	This Study	Standard Benchmark	Assessment
ROC-AUC	0.795	≥ 0.75 acceptable; ≥ 0.80 good (Hosmer & Lemeshow, 2013)	Acceptable — approaches good range
McFadden Pseudo R^2	~11.53%	$\geq 10\%$ meaningful fit (McFadden, 1974)	Acceptable
AIC	1,607.01	Lower is better (relative metric)	Used for model comparison only
Classification Threshold	0.055	Default 0.50; adjusted for class imbalance (Japkowicz & Stephen, 2002)	F1-maximised at 0.055
Event Rate	~1.31%	—	Low — threshold tuning required
n (regression sample)	~12,784	Full disbursed: 13,011 (–227 missing CIBIL)	No train-test split; policy tool only

Note: Values in this table ($AUC = 0.795$, $McFadden R^2 = 11.53\%$, $AIC = 1,607.01$) are drawn from the broader 11-variable exploratory run conducted in Excel during the analytical phase. The final Python logistic regression model (6 variables, `statsmodels.api.Logit`) produces $AUC = 0.787$, $McFadden R^2 = 10.67\%$, and $AIC = 1,612.38$ — reported in full in Chapter 5 (Table 5.6) and Annexure V. Both implementations confirm the same substantive conclusion: the model achieves acceptable discriminative ability for a credit risk screening application.

Figure 2.3 — Approximate ROC Curve (Internship Dataset — Illustrative)

True Positive Rate (Sensitivity) at Selected False Positive Rate Thresholds							
FPR	0.05	0.10	0.15	0.20	0.30	0.40	0.50
TPR	0.22	0.38	0.49	0.58	0.71	0.80	0.87
$AUC = 0.795$ · A random classifier would score $AUC = 0.50$ · No holdout validation applied							

Note: The AUC value shown here (0.795) reflects the Excel exploratory analysis. The final Python model AUC is 0.787 — see Figure 5.1 and Table 5.6 for the complete validated output.

At low event rates, the default 0.50 classification threshold would classify nearly all borrowers as non-defaulters, failing to surface at-risk accounts. Prior literature recommends threshold adjustment to balance recall and precision given the asymmetric costs of false approvals vs false rejections (Japkowicz & Stephen, 2002; He & Garcia, 2009).

This study applies a threshold of 0.055 determined by F1-score maximisation across thresholds from 0.01 to 0.50. This adjustment was necessary given the ~1.31% event rate in the dataset, and is consistent with practices recommended for imbalanced credit scoring datasets.

2.5 Rule-Based Early Warning Systems in Microfinance

Early Warning Systems (EWS) identify pre-default signals to enable intervention before accounts breach NPA thresholds. In microfinance and consumer lending, EWS typically combine bureau-based triggers, behavioural signals, and operational indicators such as auto-debit rejection rates (Schreiner, 2004; Chen & Chivakul, 2008). Prior research in Indian microfinance suggests rule-based EWS using simple threshold conditions can be operationally effective when calibrated on the institution's own portfolio data — with the advantage of being applied by collection teams without statistical tools (Ghosh, 2012; Srinivasan, 2014).

NACH failure data has been identified as a potentially useful leading indicator of NPA risk in Indian retail lending (Mahajan & Ramola, 2016). Repeated auto-debit failures are suggested to be associated with elevated DPD breach probability, though the strength of this relationship varies by product type and borrower segment (Srinivasan, 2019).

This study finds NACH failure rates in the internship dataset ranging from 7% to 11.5% across presentation cycles, with Sri Chaitanya's mid-month cycle showing the highest rate (~11.55%). These patterns are broadly consistent with the prior suggestion that NACH failure may signal NPA risk, though the single-cycle nature of the NACH data limits any stronger inference. Four rule-based Early Warning Flags (A through D) developed in Section 6.3 use NACH failure alongside bureau indicators — with default rate lift and recall reported for each flag within the dataset.

Flag	Trigger Conditions	Approx. Default Rate	Lift over Portfolio Avg.
Flag A	CIBIL < 700	~3.80%	~2.9x
Flag B	Self-Employed + Tenure = 11 EMI	~5.12%	~3.9x
Flag C	Manual Approval + CIBIL < 700	~5.56%	~4.2x
Flag D	CIBIL < 700 + Overdue > 0 + Self-Employed	~5.89%	~4.5x

Table 2.2 — Early Warning Flag Summary

2.6 RBI Guidelines on NPA Classification for NBFCs

RBI's IRAC norms govern NPA classification across all NBFCs. Under the Scale-Based Regulation framework (RBI, October 2023), any term loan overdue by more than 90 days is classified as NPA — a norm extended to all NBFCs with full compliance required by March 2025 (RBI, 2024). The framework also introduced SMA tiering: SMA-0 (1–30 DPD), SMA-1 (31–60 DPD), SMA-2 (61–90 DPD). The key provisioning and upgradation rules are summarised below.

Regulatory Item	Provision (Source)	Relevance to This Study
NPA Definition	Principal/interest overdue > 90 days (RBI, 2024)	Definitional basis for Default_Flag
90-Day Norm	Extended to all NBFCs; compliance by March 2025 (RBI, 2023)	All NBFC partners in dataset subject to this norm
SMA Classification	SMA-0: 1–30 DPD • SMA-1: 31–60 DPD • SMA-2: 61–90 DPD (RBI, 2021)	DPD bands in this study broadly align with SMA tiers
Provisioning Norms	Standard assets: 0.40% provisioning for Middle Layer NBFCs (RBI, 2024)	Informs business cost of NPA in Section 6.2
NPA Upgradation	NPA → 'Standard' only on full clearance of ALL arrears — not merely DPD < 90 (RBI, 2021)	Partial payments insufficient; shapes collection strategy (Chapter 6)

Table 2.3 — RBI NPA Regulatory Framework & Relevance to Study Design

This study uses 90+ DPD as the NPA definition throughout, consistent with IRAC norms. The SMA tiers directly shaped the DPD analysis in Chapter 6: 55% of NPA accounts sit in the 30–60 DPD band (SMA-2), which means more than half are technically still recoverable. That is what the EWS in Chapter 6 is designed to act on — not confirmed NPAs, but the 55% that haven't crossed 90 days yet.

2.7 Research Gap and Study Contributions

The literature reviewed covers the right concepts — credit risk in NBFCs, income as a default driver, CIBIL score validity, logistic regression design, rule-based EWS, and RBI provisioning norms. What is largely absent is any of it applied to short-tenure, unsecured school fee EMI products. Most Indian education loan research targets higher education or overseas study — longer-tenure, collateral-backed products with fundamentally different risk profiles (Nair & Ferdows, 2019; Nageswara Rao & Murthy, 2020). EWS work in India concentrates on microfinance and agricultural credit. The urban school fee EMI space is effectively uncovered.

Study Contribution	Description
1. Segment Application	Applies logistic regression credit scoring to a short-tenure, unsecured fee EMI portfolio — a product segment that appears underrepresented in published Indian lending research
2. Driver Findings	Finds that income type and repayment tenure are stronger predictors of default than CIBIL score alone in the dataset — consistent with, but not identical to, prior findings on retail loan defaults
3. Operational Scorecard	Develops a 4-dimension Credit Risk Scorecard calibrated on observed default rates from ~24,626 internship records, designed to be directly usable in credit operations
4. EWS Integration	Incorporates NACH failure data as an operational EWS trigger alongside bureau flags — linking collection performance data to pre-disbursement underwriting risk in a manner not commonly documented for this product type

Table 2.4 — Study Contributions to Existing Literature

These contributions come with clear limits: two institutions, one observation window, no macro controls, no holdout validation. They are operational inputs, not peer-reviewed empirical findings. But they fill a specific gap — and the four outputs (scorecard, model, EWS, simulation) are directly deployable.

CHAPTER 3 — DATA OVERVIEW & DESCRIPTIVE ANALYSIS

The data in this chapter tells a consistent story: a healthy average NPA rate of 1.34% with a deeply unhealthy distribution underneath it. Manual approvals account for 74.7% of all defaults. Eleven-EMI loans account for 54.6%. GQ_Defaulter profiles — 2% of the portfolio — default at 7.36%. The sections below build that picture variable by variable.

3.1 Dataset Description

Three datasets underpin the analysis. The key structural detail: 174 unique customer IDs are flagged as defaults — the NPA file shows 181 rows only because 7 customers hold loans at both institutions and appear twice.

Dataset	Records	Key Variables / Notes
Master_Raw_Data	24,626 rows, ~48 columns	Applicant demographics, CIBIL score, income type, loan amount, tenure, approval type, NBFC partner, disbursement status, customer profile label
NPA_Master_Data	181 rows (174 unique customers)	DPD, outstanding amount, repayment status — 7 customers hold loans at both institutions and appear twice; 174 used for all rate calculations
NACH_Summary	6 records	NACH presentation cycle data for November 2025 — success, failure, and reject counts by institution and cycle date

Table 3.1 — Internship Dataset Structure

The Default_Flag variable (binary: 1 = NPA, 0 = performing) was constructed by cross-referencing customer IDs across the Master_Raw_Data and NPA_Master_Data. A total of 174 unique customer IDs were flagged as defaults and used for all NPA rate calculations throughout this report. The figure of 181 appears in the NPA_Master_Data because 7 customers hold active loans at both Narayana and Sri Chaitanya and are recorded in both institution-level NPA files; both figures are correct within their respective analytical contexts.

3.2 Application Funnel Analysis

Of 24,626 total applications received, 13,011 (53%) reached the disbursed stage. The remaining applications were rejected, backed out, or classified as duplicates at various stages of the credit and execution pipeline. The funnel breakdown is presented below.

Funnel Stage	Count	% of Total	Avg Loan Amt (₹)	Key Observation
Loan Live	9,400	38.2%	₹85,911	Active loans — largest single stage
Loan Closed	3,611	14.7%	₹92,411	Successfully repaid — healthy portfolio outcome
Duplicate	4,541	18.4%	₹73,928	⚠ Data quality concern — must be excluded from all rate calculations
Rejected	3,790	15.4%	₹76,601	Filtered by credit team prior to disbursement
Post-Credit Back Out	1,907	7.7%	₹76,285	Approved but customer withdrew post-credit decision
Pre-Credit Back Out	1,062	4.3%	₹89,991	Customer withdrew before credit decision — documentation/process drop
LE+ Stage	272	1.1%	₹46,795	Loan Execution stage — near disbursement
CR/ID & Drop Out	43	0.2%	—	Credit review / minimal drop-outs

Table 3.2 — Application Funnel Stage Distribution [Excel: Analysis 1.2]

The credit team is filtering: 15.4% outright rejected. But the 18.4% duplicate rate means nearly 1 in 5 records cannot be used in analysis without prior cleaning — a data governance gap that should be closed.

3.3 Overall NPA Rate

Narayana (1.32%) and Sri Chaitanya (1.35%) are nearly identical at the headline level. That similarity is the least interesting fact about the data — what matters is where the defaults are coming from.

Metric	Narayana	Sri Chaitanya	Combined
Total Applications	9,463	15,163	24,626
Disbursed Loans	4,917	8,094	13,011
Defaults (NPAs)	65	109	174
NPA Rate (%)	1.32%	1.35%	1.34%
Rejected / Backed Out	4,546	7,069	11,615

Table 3.3 — Overall NPA Rate by Institution [Excel: Analysis 1.1]

The combined NPA rate of 1.34% appears comparatively contained relative to broader NBFC education lending estimates of 2–5%. Narayana (1.32%) and Sri Chaitanya (1.35%) show near-identical default rates despite differing application volumes, suggesting similar risk profiles across the two institutional partners in the dataset. Notably, 11,615 applications (47%) were rejected or backed out, indicating a substantial volume of higher-risk applicants entering but not completing the funnel.

1.34% overall. Benchmark: 2–5%. The average is fine. The segments underneath it are not.

3.4 Default Distribution by Key Variables

Five variables drive the variation in NPA rates. Each one tells a distinct story.

3.4.1 CIBIL Score Band

CIBIL Band	Disbursed Loans	Default Count	Default Rate	% of All Defaults	Risk Level
> 750	9,029	66	0.73%	37.9%	Low
700–750	3,103	73	2.35%	42.0%	Moderate
650–700	647	29	4.48%	16.7%	High
NTC	227	6	2.64%	3.4%	Moderate-High
< 650	4	0	—	—	Very High (insuf. data)

Table 3.4 — Default Rate by CIBIL Score Band [Excel: Analysis 1.3]

The 650–700 band has the worst rate (4.48%) but the 700–750 band generates the most defaults in absolute terms (73 of 174) — it is the largest moderate-risk group in the portfolio. NTC borrowers at 2.64% carry more risk than their thin bureau history suggests. CIBIL score is necessary but not sufficient: a 650–700 self-employed borrower defaults at 7.14% while a salaried > 750 borrower defaults at 0.56% — the same CIBIL range, eleven times the outcome difference.

Figure 3.1 — Default Rate by CIBIL Band (Internship Dataset)

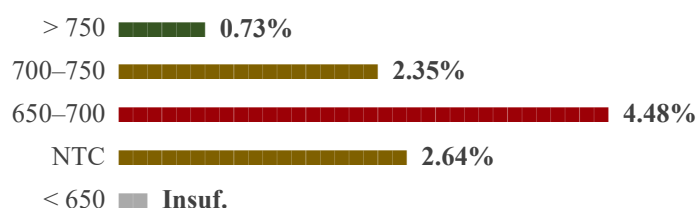


Figure 3.1 — Default rate by CIBIL band (internship dataset; directional)

3.4.2 Income Type

Income Type	Disbursed Loans	% of Disbursed	Default Count	Default Rate	% of All Defaults
Salaried	10,115	77.7%	96	0.95%	55.2%
Self-Employed	2,896	22.3%	78	2.69%	44.8%

Table 3.5 — Default Rate by Income Type [Excel: Analysis 1.4]

Self-employed borrowers are 22.3% of the portfolio but 44.8% of defaults. At 2.69% versus 0.95% for salaried, the gap is statistically significant ($p < 0.001$, Z-test) and operationally significant: nearly half of all NPAs come from less than a quarter of borrowers.

3.4.3 Repayment Tenure

Tenure	Disbursed Count	% of Disbursed	Default Count	Default Rate	% of All Defaults
6 EMIs	6,547	50.3%	28	0.43%	16.1%
9 EMIs	2,579	19.8%	51	1.98%	29.3%
11 EMIs	3,885	29.9%	95	2.45%	54.6%

Table 3.6 — Default Rate by Repayment Tenure [Excel: Analysis 1.8]

The 11-EMI product is 29.9% of disbursements and 54.6% of defaults. At 2.45% versus 0.43% for 6-EMI, it defaults at nearly six times the rate. This is the strongest predictor in the regression model (OR = 3.83) and the most straightforward policy fix: the product itself is the risk, not just the borrower.

Figure 3.2 — Default Rate by Tenure (Internship Dataset)

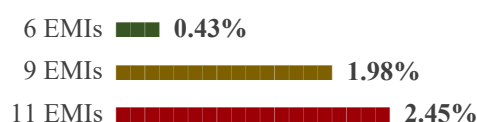


Figure 3.2 — Default rate by tenure (internship dataset; directional)

Figure 3.3 — Heatmap: Default Rate by Tenure × Approval Channel

Tenure (Avg Rate)	Bureau Approval (Automated)	Manual Approval (Override)	Manual:Bureau Ratio	Risk Signal
6-EMI (0.43% avg)	0.19% <i>n</i> = 4,120	0.84% <i>n</i> = 2,255	4.4x	✓ Low
9-EMI (1.98% avg)	1.12% <i>n</i> = 1,337	3.00% <i>n</i> = 1,200	2.7x	⚠ High
11-EMI (2.45% avg)	1.13% <i>n</i> = 1,775	3.63% <i>n</i> = 2,065	3.2x	⚠ High

Colour scale: ■ White < 0.5% ■ Yellow 0.5–1.0% ■ Orange 1.0–2.0% ■ Dark Orange 2.0–3.0% ■ Red 3.0–4.0% ■ Deep Red > 4.0%

Figure 3.3 — Heatmap: Default rate by Tenure × Approval Channel (internship dataset) | Cell shade = default rate intensity

The best combination in the portfolio: 6-EMI + Bureau at 0.19%. The worst: 11-EMI + Manual at 3.63% — 19 times higher. Bureau approvals perform consistently across all three tenures (0.19%, 1.12%, 1.13%). Manual approvals fall apart as tenure extends (0.84%, 3.00%, 3.63%). The 11-EMI manual combination alone generates 75 defaults from 2,065 loans. No other single product-channel combination comes close.

3.4.4 Approval Channel

Approval Type	Disbursed Count	% of Disbursed	Default Count	Default Rate	% of All Defaults	Risk Signal
Manual-NBFC-Assigned	5,520	42.4%	130	2.36%	74.7%	High — manual overrides risky
Bureau-NBFC-Assigned	7,232	55.6%	43	0.59%	24.7%	Low — automated decisions safer
Banking-NBFC-Assigned	207	1.6%	1	0.48%	0.6%	Very Low
Bureau-CIBIL-NBFC-Assigned	52	0.4%	0	0%	0%	Very Low

Table 3.7 — Default Rate by Approval Channel [Excel: Analysis 1.11]

42.4% of disbursements. 74.7% of defaults. That is the manual approval channel. At 2.36% versus the bureau's 0.59%, it is 4x riskier — and the gap exists in every CIBIL band, as Figure 3.4 shows.

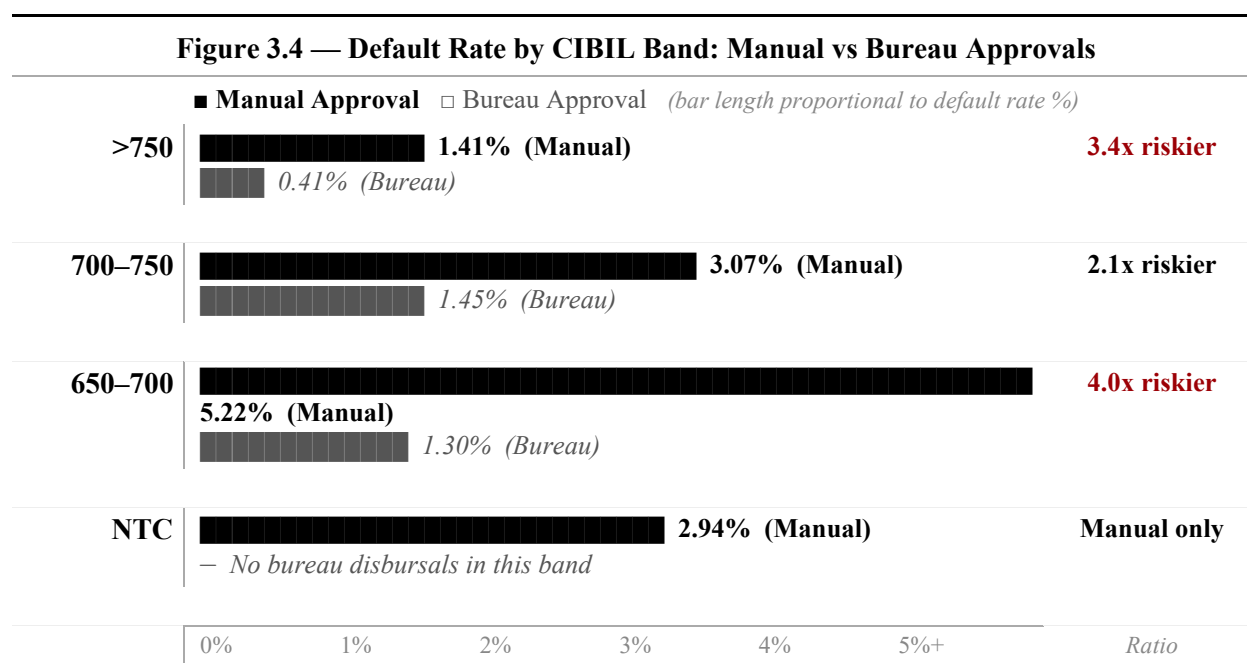


Figure 3.4 — Clustered comparison: Default rate by CIBIL band for Manual vs Bureau approvals (internship dataset)

In the 650–700 band: 5.22% manual, 1.30% bureau — 4.0x. In the > 750 band: 1.41% manual, 0.41% bureau — 3.4x. The manual channel makes even the safest CIBIL profiles significantly riskier. The bureau algorithm is working. The override process is undoing it.

3.4.5 NBFC Partner

NBFC Partner	Disbursed Loans	Total Portfolio (₹)	Default Count	Default Rate	% of All Defaults	Risk Flag
Mirae Asset Financial Svcs	4,719	₹38.52 Cr	24	0.51%	13.8%	Acceptable
Fintree Finance	310	₹2.29 Cr	2	0.65%	1.1%	Acceptable
Arka Fincap (Kirkoskar Group)	3,600	₹37.48 Cr	43	1.19%	24.7%	Acceptable
Western Capital	1,506	₹10.22 Cr	24	1.59%	13.8%	Elevated
Ratnaafin Capital Pvt Ltd	2,094	₹18.97 Cr	41	1.96%	23.6%	Elevated
Ratnaafin Capital Pvt Ltd Lite	777	₹6.52 Cr	40	5.15%	23.0%	High

Table 3.8 — Default Rate by NBFC Partner [Excel: Analysis 1.12]

Ratnaafin Lite at 5.15% is the outlier — 3.8x the portfolio average, generating 23% of all defaults from 6% of disbursements. Together, both Ratnaafin products account for 47% of all defaults. Mirae Asset, the largest partner by volume at 4,719 loans, holds its default rate to 0.51%. Product eligibility criteria and approval standards explain much of this divergence.

3.5 Customer Profile Label Distribution

The GQ_Defaulter label is the most alarming entry in the dataset: 7.36% default rate from customers who had already defaulted on a GrayQuest loan and were approved again. CIBIL_Defaulter profiles (3.98%) were still being disbursed at over 1,000 loans. Together these two high-risk labels represent 11.1% of the portfolio but 38% of all defaults.

Profile Label	Disbursed	% of Portfolio	Default Rate	Risk Classification
GQ_Defaulter	258	2.0%	7.36%	Critical
age not within policy norms	101	0.8%	5.94%	High
CIBIL_Defaulter	1,182	9.1%	3.98%	High
NTC	223	1.7%	2.69%	Moderate-High
Good_Cibil_but_OD	130	1.0%	2.31%	Moderate
Low_income	313	2.4%	2.24%	Moderate
Other	280	2.2%	2.14%	Moderate
No_Positive_acc	455	3.5%	1.32%	Moderate
CIBIL_Defaulter_Settled	76	0.6%	1.32%	Moderate
Good_Cibil	9,574	73.6%	0.74%	Low
Thin_Cibil	338	2.6%	0.59%	Low

Table 3.9 — Customer Profile Label Default Rates [Excel: Analysis 1.10] | Red = Critical/High • Yellow = Moderate • Green = Low

Good_Cibil contributes 41% of defaults — but only because it is 73.6% of the portfolio. Its 0.74% rate is healthy. The scorecard's job is to stop the 11.1% of GQ_Defaulter and CIBIL_Defaulter profiles from contributing 38% of the damage.

3.6 Geographic Distribution

Geography is a useful lens for collection prioritisation, but not for underwriting — broad region doesn't predict default. State-level data tells a more useful story.

Geography	Disbursed Loans	Default Count	Default Rate	% of All Defaults	Observation
AP/TL (home market)	6,812	91	1.34%	52.3%	Highest volume — home market
Non-AP/TL	6,199	83	1.34%	47.7%	Similar default rate to home market
— TG (Telangana)	2,475	40	1.62%	23.0%	Above average — monitor closely
— MH (Maharashtra)	1,815	18	0.99%	10.3%	Below average — best major state
— AP (Andhra Pradesh)	1,486	17	1.14%	9.8%	Near average
— RJ (Rajasthan)	104	3	2.88%	1.7%	Highest rate — small sample, caution

Table 3.10 — Default Rate by Geography [Excel: Analysis 1.9]

AP/TL and Non-AP/TL are identical at 1.34% — geography does not predict default at the regional level. At the state level: Telangana (1.62%) and Rajasthan (2.88%) are above average; Maharashtra (0.99%) is the best major state. Rajasthan's rate is based on only 104 loans — treat it as a flag, not a finding. The practical use of this table is collection resource allocation, not underwriting.

3.7 DPD Severity Distribution of NPA Accounts

More than half the NPA book — 55.2%, or 100 accounts — is in the 30–60 DPD band. These accounts have not yet crystallised into confirmed NPAs. They are recoverable. And they carry the highest average outstanding at ₹59,954.

DPD Band	NPA Count	% of All NPAs	Avg Max DPD	Total Outstanding (₹)	Avg Outstanding (₹)	Severity
30–60 DPD	100	55.2%	48 days	₹59.95 L	₹59,954	Early Stage — Recoverable
60–90 DPD	45	24.9%	75 days	₹25.70 L	₹57,107	Serious — Intervention Needed
90+ DPD	36	19.9%	132 days	₹17.54 L	₹48,712	Critical — Write-off Risk
TOTAL	181	100%	—	₹1.03 Cr	—	—

Table 3.11 — DPD Severity Distribution of NPA Accounts [Excel: Analysis 1.14a]

The 90+ DPD accounts (19.9%) are largely lost — 132 days average overdue, ₹48,712 average outstanding. The priority window is the 30–60 DPD band: ₹59.95 lakh in outstanding, still early, still collectible. Sri Chaitanya's 90+ DPD share (25.4%) is nearly 2.5x Narayana's (10.4%) — the same pattern that shows up in SC's worse NACH failure rates. Total NPA outstanding: ₹1.03 crore.

3.8 NACH Collection Performance

In November 2025, GrayQuest ran 11,143 NACH debit attempts across both institutions. 910 failed — 8.2% of all presentations. That is 910 borrowers who did not have enough in their account when the EMI was due. Every one of them is a potential DPD case in the making.

Institution	Cycle	Date	Total Presented	Success	Failed	Success Rate	Failure Rate
Narayana	1st Presentation	04-Nov-25	641	595	45	92.8%	7.0%
Narayana	2nd Presentation	05-Nov-25	2,172	1,970	196	90.7%	9.0%
Narayana	Mid-Month	10-Nov-25	1,115	1,003	110	89.9%	9.9%
Sri Chaitanya	1st Presentation	04-Nov-25	570	528	42	92.6%	7.4%
Sri Chaitanya	2nd Presentation	05-Nov-25	4,190	3,837	349	91.6%	8.3%
Sri Chaitanya	Mid-Month	10-Nov-25	1,455	1,286	168	88.4%	11.6%
COMBINED	All Cycles	Nov-25	11,143	—	910	~91.8%	~8.2%

Table 3.12 — NACH Collection Performance [Excel: Analysis 1.13]

Sri Chaitanya's mid-month cycle failed at 11.6% — 168 out of 1,455 attempts on the 10th of November, none of them paid. That cycle is the single worst collection data point in the dataset, and SC's 90+ DPD count (29 versus Narayana's 7) is the downstream consequence. Failure rates trend upward through the month for both institutions: by mid-month, the customers who were going to struggle have already missed two debit attempts. That is the intervention window.

CHAPTER 4 — RISK SEGMENTATION & STATISTICAL VALIDATION

The scorecard built in this chapter assigns every applicant a risk score from 0 to 9 across four observable dimensions — CIBIL band, income type, tenure, and approval channel. Those four dimensions are not arbitrary: each one is statistically significant, empirically validated, and operationally actionable.

4.1 Multi-Dimensional Risk Segmentation

Four variables. Each one independently predicts default. Combined, they produce a 0–9 point score that separates borrowers into four tiers with dramatically different default rates.

Applied to 13,011 disbursed records:

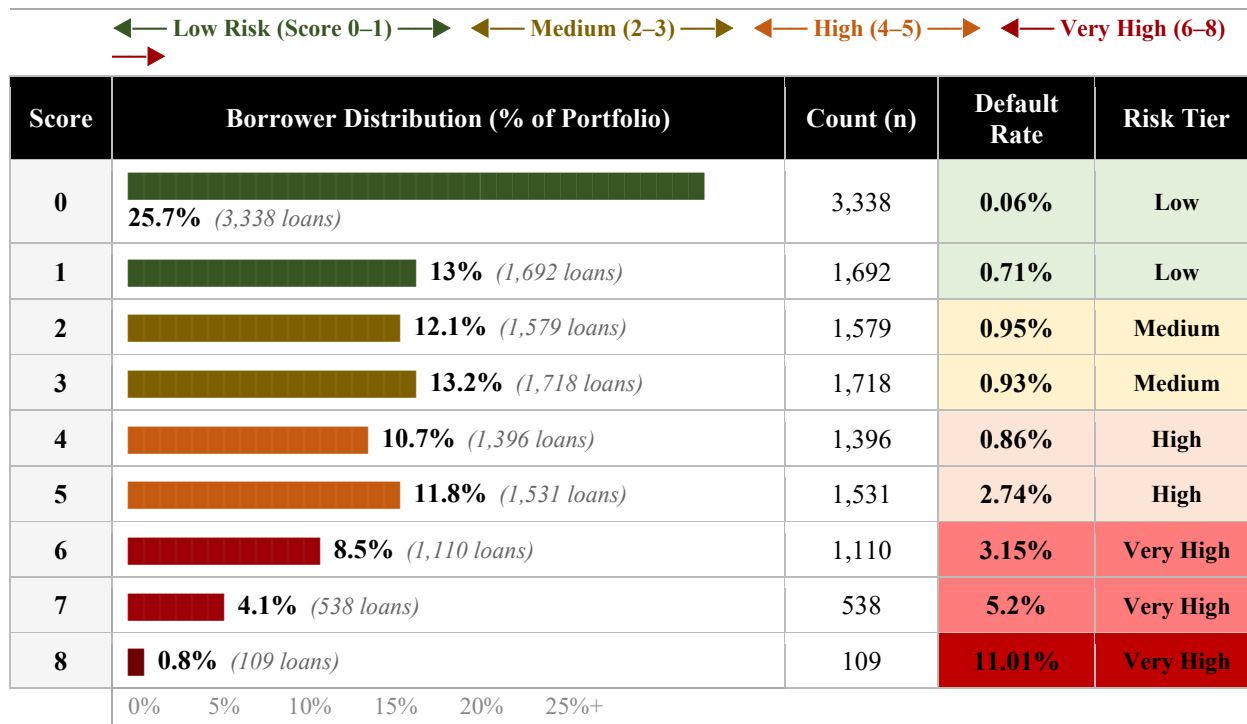
Risk Tier	Disbursed Count	% of Portfolio	Default Count	Default Rate	Avg CIBIL	Avg Overdue (₹)	Recommended Action
Low Risk	9,425	72.4%	62	0.66%	775	₹15	Instant Approval
Medium Risk	2,358	18.1%	49	2.08%	743	₹1,774	Conditional Approval
High Risk	1,228	9.4%	63	5.13%	711	₹25,837	Reject / Strict Review

Table 4.2 — Risk Tier Results: Disbursed Portfolio [Excel: Analysis 2.1]

High Risk (9.4%) defaults at 5.13% with an average bureau overdue of ₹25,837. Low Risk (72.4%) defaults at 0.66% with an average overdue of ₹15. The credit base is sound. The 9.4% is the problem.

Figure 4.2 — Histogram: Borrower Distribution Across Risk Score Bands (0–8)

Bar width $\propto$ % of portfolio | $n = 13,011$ disbursed loans | Score 0 = lowest risk • Score 8 = highest risk



Low Risk (0–1): 38.7% of portfolio | 0.28% default rate **Medium (2–3): 25.3% | 0.94%** **High (4–5): 22.5% | 1.84%** **Very High (6+): 13.5% | 4.27%**

Note: Score 5 spike (2.74% rate) and Score 8 extreme (11.01% rate) are key policy trigger points

Figure 4.2 — Histogram: Borrower distribution by risk score band (0–8) with default rate overlay (internship dataset)

38.7% of borrowers score 0 or 1. The portfolio is not broadly risky — it is concentrated risky. Watch Score 5: it jumps from 0.86% at Score 4 to 2.74%, more than tripling in one step. Score 8 reaches 11.01% — the most extreme single bucket, 109 loans, 12 defaults.

The jump at Score 5 is the policy threshold that matters. Below it, the model can calibrate between approval and conditional approval. Above it, the risk escalates non-linearly — which is exactly why Score ≥ 6 is the correct auto-reject boundary, not a smooth extension of the lower tiers.

4.2 Credit Risk Scorecard

The scorecard is a lookup table. Four inputs — CIBIL band, income type, tenure, approval channel — each get a point weight calibrated on observed default rates. Add them up. The total score determines the approval response. No subjectivity, no overrides.

Section A — Point Assignment Table

#	Dimension	Category / Value	Points	Observed Default Rate	Rationale
1	CIBIL Score	> 750 (Good)	0	0.73%	Lowest risk band — strong bureau history
	CIBIL Score	700–750 (Fair)	1	2.35%	Moderate risk — manageable with monitoring
	CIBIL Score	650–700 (Weak)	2	4.48%	~6x riskier than > 750 band
	CIBIL Score	NTC (New to Credit)	2	2.64%	No bureau history — treated as moderate risk
	CIBIL Score	< 650 (Poor)	3	Insuf. data	Highest bureau risk — very few disbursals
2	Income Type	Salaried	0	0.95%	Stable, verifiable income — low volatility
	Income Type	Self-Employed	1	2.69%	2.84x riskier; income irregular and harder to verify
3	Tenure (EMI)	6-EMI	0	0.43%	Short repayment cycle — lowest default rate
	Tenure (EMI)	9-EMI	1	1.98%	Moderate — 4.6x riskier than 6-EMI
	Tenure (EMI)	11-EMI	2	2.45%	Strongest regression predictor (OR $\approx$ 3.83)
4	Approval Type	Bureau / Banking	0	0.59%	Data-driven — low override risk
	Approval Type	Manual Override	3	2.36%	4x higher default rate; 75% of all NPAs

Table 4.3 — Scorecard Point Assignment [Excel: Credit Risk Scorecard Sheet]

Maximum Possible Score = 9 pts (CIBIL < 650 [3] + Self-Employed [1] + 11-EMI [2] + Manual Approval [3])

Section B — From Score to Decision

Total Score	Risk Category	% of Portfolio	Observed Default Rate	Risk Interpretation
0 – 1	Low Risk	72.4%	0.66%	Strong credit profile — auto-approve eligible
2 – 3	Medium Risk	18.1%	2.08%	Moderate risk — conditional approval with documentation
4 – 5	High Risk	9.4%	5.13%	High default likelihood — senior review needed
6 – 9	Very High Risk	5.9%	6.04%	Extreme risk — auto-reject; no manual overrides

Table 4.4 — Score to Risk Category Mapping

Figure 4.3 — Default Rate Escalation Across Risk Tiers: Low → Very High

Bar length $\propto$ default rate | $n = 13,011$ disbursed loans | Colour intensity reflects risk level

Risk Tier	Default Rate (Bar Chart)	% of Portfolio	% of All Defaults	Avg CIBIL
Low Risk (Score 0–1)	0.28% <i>Baseline</i> 14 defaults	38.7% (5,030)	8% <i>of all NPAs</i>	780
Medium Risk (Score 2–3)	0.94% <i>3.4x Low Risk</i> 31 defaults	25.3% (3,297)	17.8% <i>of all NPAs</i>	767
High Risk (Score 4–5)	1.84% <i>6.6x Low Risk</i> 54 defaults	22.5% (2,927)	31% <i>of all NPAs</i>	755
Very High Risk (Score 6+)	4.27% <i>15.2x Low Risk</i> 75 defaults	13.5% (1,757)	43.1% <i>of all NPAs</i>	726

Default Rate: 0.28% → 0.94% → 1.84% → **4.27%** (15.3x increase from Low to Very High Risk)

Portfolio share vs NPA share: Low Risk 38.7% portfolio → only 8.0% of NPAs | **Very High 13.5% portfolio → 43.1% of NPAs**

Figure 4.3 — Default rate escalation from Low to Very High Risk tier (internship dataset)

15.3x from Low to Very High Risk. The steepest jump is at the top: High Risk 1.84% → Very High 4.27%. A single credit threshold would miss this entirely. Four tiers are not bureaucratic complexity — they reflect four genuinely different risk profiles.

The inverse relationship does the persuading: 38.7% of borrowers, 8.0% of NPAs (Low Risk). 13.5% of borrowers, 43.1% of NPAs (Very High). Auto-approve the first group. Reject the second.

Section C — Final Approval Decision Table

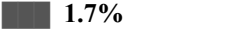
Score	Risk Category	Decision	Key Conditions	Expected NPA Rate
0–1	Low Risk	APPROVE	Auto-approve; fast-track disbursement; no additional documentation required	~0.66%
2–3	Medium Risk	CONDITIONAL APPROVE	Co-applicant or guarantor; income proof (ITR/GST); loan cap at ₹75,000	~2.08%
4–5	High Risk	ENHANCED REVIEW	Senior credit officer sign-off; CIBIL re-verification; income floor check (min 5x annual EMI); 11-EMI product restricted	~5.13%
6–9	Very High Risk	REJECT	Auto-reject; no manual override; GQ Defaulter & CIBIL Defaulter profiles included — saves ~₹12.1L NPA/cycle	~6.04%

Table 4.5 — Tiered Approval Decision Framework

Figure 4.1 — Pareto Chart: Cumulative NPA Contribution by Risk Segment

Segments sorted by default count (descending) | n = 174 total defaults | Bar = % of all NPAs | Line = Cumulative %

Risk Segment	NPA Bar (% of All Defaults)	Defaults (n / %NPA)	Cumulative NPA %	Default Rate
CIBIL_Defaulter + Manual ⚠️ Avoidable	25.3%	44 (25.3%)	25.3% ◀ Top segment	4.44%
All Others ✓ Structural	21.8%	38 (21.8%)	47.1% ◀ Top segment	0.51%
11-EMI + Manual (non-Defaulter) ⚠️ Avoidable	12.1%	21 (12.1%)	59.2% ◀ 80% line	3.57%
GQ_Defaulter (any channel) ⚠️ Avoidable	10.9%	19 (10.9%)	70.1% ◀ 80% line	7.36%

Good CIBIL + 11-EMI + Manual <i>⚠ Avoidable</i>	 10.3%	18 (10.3%)	80.4% ◀ 95% line	1.96%
Good CIBIL + 11-EMI + Bureau <i>✓ Structural</i>	 9.8%	17 (9.8%)	90.2% ◀ 95% line	1.04%
9-EMI + Manual (non-Defaulter) <i>⚠ Avoidable</i>	 8%	14 (8%)	98.2%	1.55%
CIBIL_Defaulter + Bureau <i>✓ Structural</i>	 1.7%	3 (1.7%)	99.9%	1.58%

80% of NPAs come from just 4 segments | These 4 segments = only 21.2% of the portfolio | *⚠ Avoidable = addressable through policy intervention*

Cumulative % fill: ■ Orange = top contributors ($\leq 50\%$) ■ Red = 50–80% ■ Grey = 80–100% Bar colour: ■ Red = avoidable □ Grey = structural

Figure 4.1 — Pareto Chart: Cumulative NPA Contribution by Risk Segment (internship dataset, $n=174$ defaults)

80% of NPAs from 21.2% of the portfolio. The top single segment — CIBIL_Defaulter customers approved manually — generates 25.3% of all defaults from 7.6% of disburseals. GQ_Defaulter is 2.0% of the portfolio and 10.9% of defaults at 7.36%. These are not statistical artefacts. They are names in a database with a known risk profile who are still being approved.

Three interventions cover the five avoidable segments: auto-reject GQ_Defaulter and CIBIL_Defaulter on manual, restrict 11-EMI to non-manual channels. The structural segments — All Others and Good CIBIL + Bureau — will always generate some defaults. That is normal. The 0.91% NPA target requires removing the avoidable five from the pool, not chasing the structural two.

4.3 CIBIL × Income Type Cross-Segmentation

Salaried + CIBIL > 750: 0.56% default rate. Self-Employed + CIBIL 650–700: 7.14%. Both groups might pass a basic CIBIL screen. One defaults at 12.7x the rate of the other. That is why income type is a separate scoring dimension.

CIBIL Band	Income Type	Disbursed	Default Count	Default Rate	Risk Rank	Approval Guidance
650–700	Self-Employed	196	14	7.14%	#1 — Highest	Reject / Strict Review
NTC	Self-Employed	67	4	5.97%	#2	Reject (NTC + SE = Max Risk)
700–750	Self-Employed	981	35	3.57%	#3	Conditional — Strong docs only
650–700	Salaried	451	15	3.33%	#4	Conditional — Salary proof required
700–750	Salaried	2,122	38	1.79%	#5	Conditional Approval
> 750	Self-Employed	1,651	25	1.51%	#6	Approve with business income proof
NTC	Salaried	160	2	1.25%	#7	Conditional — EPFO/salary required
> 750	Salaried	7,378	41	0.56%	#8 — Lowest	Instant Approval eligible

Table 4.7 — CIBIL × Income Type Risk Matrix [Excel: Analysis 2.2]

Even in the > 750 band — the 'safe' tier — self-employed borrowers default at 1.51% versus 0.56% for salaried. Income type is not a tiebreaker. It is a primary variable. The 700–750 self-employed combination (3.57%) is riskier than the 650–700 salaried combination (3.33%) — a reversal that no CIBIL-only model would catch.

4.4 Additional Segmentation Findings

4.4.1 HSA Bucket Analysis

HSA does not predict default — default rates across HSA buckets run from 1.19% to 1.89% with no clear direction. The only meaningful HSA signal is the NTC category (3.33%). HSA may be useful for credit limit sizing but should not be a standalone approval criterion.

4.4.2 Student Type: New vs Existing

New customers and renewal customers default at exactly the same rate: 1.34% each. Renewal status is not a risk signal. Every returning customer should be re-scored from scratch — their CIBIL and income profile may have changed, and the historical relationship offers no predictive protection.

4.4.3 Funnel Leakage Analysis

The rejection filter is working — 76% of rejected applications are High Risk. But 1,236 High Risk applicants still made it through to disbursement (13.1% of Loan Live). Of those, GQ_Defaulter and CIBIL_Defaulter profiles alone generated 66 of 174 observed defaults (38%) from 11.1% of the portfolio. The filter has a leak, and the leak is expensive.

4.5 Institution Comparison: Narayana vs Sri Chaitanya

Metric	Narayana	Sri Chaitanya	Difference / Signal
NPA Rate	1.32%	1.35%	Near-identical — similar systemic risk
Avg CIBIL Score	763.2	764.5	Virtually identical — good overall quality
Avg Loan Amount	₹1,11,560	₹73,229	Narayana loans ~52% higher on average
Self-Employed %	23.0%	21.8%	Similar SE exposure across both
Manual Approval %	43.8%	41.6%	Both show high manual approval share
11-EMI Default Rate	2.10%	2.73%	SC 11-EMI ~30% worse than Narayana
NTC Default Rate	0.00%	3.80%	SC has riskier NTC approvals — significant gap
90+ DPD Cases	7	29	SC has 4x more critical NPA cases — weaker collections

Table 4.8 — Institution-Level Comparative Segmentation [Excel: Analysis 2.8]

Sri Chaitanya's NTC borrowers default at 3.80%. Narayana's default at 0%. Same product, same policy, different applicant mix and approval quality. SC also has 29 accounts in 90+ DPD versus Narayana's 7 — four times as many critical cases. The headline NPA rates (1.32% vs 1.35%) are nearly identical, which is why segment-level comparison matters: the similarity at the top masks a material difference underneath.

4.6 Hypothesis Testing

Four of the five tests reject the null hypothesis at $p < 0.001$ — the strongest possible statistical confirmation for risk drivers already visible in the data. The one test that does not reject (loan amount) is equally important: it means loan size can be removed from every credit rule GrayQuest currently uses.

Test	Hypothesis	Method	Key Statistics	Decision	Plain Language Finding
H1: CIBIL Score	H0: No difference in mean CIBIL between defaulters and non-defaulters	Welch's t-test (independent samples)	$t = -8.92$ $p < 0.001$ Cohen's $d = 0.73$ Mean diff = 24.79 pts (764.3 vs 739.5)	REJECT H0	Defaulters have significantly lower CIBIL scores. Difference is real, not random. Cohen's $d = 0.73$ indicates medium-large practical effect.
H2: Income Type	H0: Default rate of SE = default rate of Salaried	Z-test for two proportions (one-tailed)	$Z = 7.21$ $p < 0.001$ Relative Risk = 2.84x Rate diff = 1.74 pp	REJECT H0	SE borrowers are 2.84x more likely to default. Statistically proven beyond doubt.
H3: Loan Amount	H0: No difference in mean loan amount between defaulters and non-defaulters	Welch's t-test (independent samples)	$t = 0.29$ $p = 0.774$ Mean diff = ₹1,259 (negligible)	FAIL TO REJECT H0	Loan amount does NOT predict default. Defaulters and non-defaulters borrow virtually the same amount.
H5: CIBIL Buckets	H0: Default rates equal across all CIBIL buckets	One-way ANOVA + Chi-square + Post-hoc Bonferroni pairwise	ANOVA: $F = 13,632$, $p < 0.001$ Chi-sq = 100.88, $p < 0.001$ Post-hoc: 700–750 and NTC statistically similar ($p = 0.96$)	REJECT H0	Default rates differ significantly across CIBIL buckets. 650–700 is 6x riskier than > 750. 700–750 and NTC are statistically similar risk.
H7: Approval Channel	H0: Default rate of Manual = default rate of Bureau approvals	Z-test for two proportions (one-tailed)	$Z = 8.51$ $p < 0.001$ Manual: 2.36% vs Bureau: 0.59% 75% of all defaults from manual	REJECT H0	Manual approvals are 4x riskier than bureau approvals. This is the single most operationally significant finding.

Table 4.9 — Hypothesis Testing Summary [Excel: Hypothesis Testing Sheet — Analyses 3.1, 3.2, 3.3, 3.5, 3.7]

The post-hoc test on CIBIL buckets has one particularly useful finding: the 700–750 and NTC bands are statistically indistinguishable ($p = 0.96$). That means NTC borrowers should be treated the same as 700–750 in the approval process — not as a separate, lighter-touch category.

4.7 Correlation & Driver Analysis

Individual correlations with default are small — $r < 0.10$ across the board. That is not a problem; it is expected when a single variable is trying to predict a multi-causal outcome. The ranking is what matters: which variables carry signal and which carry noise.

4.7.1 Individual Variable Correlations with Default_Flag

Rank	Variable	Spearman r	Direction	Significant?	Key Finding
1	Total_DPD_30plus_1yrs	+0.097	Risk factor (+)	YES *** $p < 0.001$	Past 30+ DPD history — strongest individual predictor; 42.5% of defaulters had this flag
2	CIBIL_Score	-0.081	Protective (-)	YES *** $p < 0.001$	Higher CIBIL → lower default; core screening metric
3	Total_Amt_Overdue_all	+0.081	Risk factor (+)	YES *** $p < 0.001$	Any existing overdue in bureau — 35.6% of defaulters had overdue vs 12.9% of non-defaulters
4	Total_DPD_90plus_acc	+0.074	Risk factor (+)	YES *** $p < 0.001$	No. of accounts with 90+ DPD — 20.7% of defaulters vs 5.6% non-defaulters
5	annual_income	-0.052	Protective (-)	YES *** $p < 0.001$	Lower income → higher default; works better with log-transform
6	Total_CC_Accounts	-0.040	Protective (-)	YES * $p < 0.05$	More credit cards = better credit profile — weak but directional
7	Total_DPD_90plus_1yrs	+0.004	—	NO (ns) $p = 0.68$	Raw 90+ DPD days — NOT a reliable predictor; excluded from model
8	loan_amount	-0.003	—	NO (ns) $p = 0.77$	Zero predictive power — confirmed non-predictor; excluded

Table 4.10 — Variable Correlation Rankings with Default_Flag [Excel: Analysis 4.8]

Past 30+ DPD history ($r = +0.097$) is the single strongest individual signal — 42.5% of defaulters had this flag. CIBIL score ($r = -0.081$) and existing overdue ($r = +0.081$) follow. Loan amount ($p = 0.77$) and raw 90+ DPD days ($p = 0.68$) contribute nothing. Both excluded.

4.7.2 Multicollinearity Assessment

DPD_30plus_1yrs is retained over DPD_90plus_acc — same concept, stronger signal. Current balance is out — it correlates with income at $r = 0.46$ and adds nothing independent. All six retained variables passed VIF checks.

CHAPTER 5 — PREDICTIVE MODELLING: LOGISTIC REGRESSION

5.1 Model Specification & Variable Selection

Dependent Variable

Default_Flag (binary: 1 = NPA / defaulted customer, 0 = performing). Constructed by cross-referencing customer IDs in Master_Raw_Data against NPA_Master_Data. The regression sample contains 168 defaults out of 12,784 disbursed records — 6 fewer than the full dataset's 174 defaults because those 6 defaulters also had missing CIBIL scores and were excluded along with their corresponding records. The event rate in the regression sample is 1.31%.

Variable Selection Rationale

Two variables excluded outright:

- Loan amount: non-significant in both hypothesis testing ($p = 0.774$) and correlation analysis (Spearman $r = -0.003$, $p = 0.77$), with near-zero predictive power confirmed in the regression ($p = 0.917$, OR = 1.008).
- Raw Total_DPD_90plus_1yrs: not significant ($p = 0.68$) in correlation analysis; Total_DPD_90plus_acc (number of delinquent accounts) is the more reliable DPD signal. Both DPD variables were also found to be highly collinear with overdue amount ($r = 0.47$ – 0.67) and were absorbed into the overdue variables in the combined model.

Feature Engineering

Feature	Source Variable	Transformation	Rationale
CIBIL_norm	CIBIL_Score	Z-score: $(x - 763) / 32.5$	Standardised so coefficient is comparable to binary features; confirmed predictor (H1: $p < 0.001$)
SE_Flag	income_type	Binary dummy: 1 = Self-Employed, 0 = Salaried	Captures income instability; reference = Salaried; validated (H2: $Z = 7.21$, $p < 0.001$)
Manual_Flag	Approval_type	Binary dummy: 1 = manual-nbfc-assigned, 0 = bureau	Manual overrides = 4x higher default rate; validated (H7: $Z = 8.51$, $p < 0.001$)
Tenure_9	tenure	Binary dummy: 1 = 9-EMI, 0 otherwise	Reference = 6-EMI (lowest-risk tenure, 0.43% default rate)

Feature	Source Variable	Transformation	Rationale
Tenure_11	tenure	Binary dummy: 1 = 11-EMI, 0 otherwise	11-EMI descriptively shows 2.45% default rate — ~6x the 6-EMI baseline
Overdue_Flag	Total_Amt_Overdue_all	Binary dummy: 1 = any overdue > ₹0, 0 otherwise	35.6% of defaulters had overdue vs 12.9% of non-defaulters (Section 4.7); binary flag more stable than continuous overdue

Table 5.1 — Feature Engineering Summary [Python: GrayQuest_Risk_Model.ipynb]

5.2 Python Implementation

Full code in Annexure V. Key implementation parameters:

Component	Detail
Libraries Used	pandas — data loading, cleaning, feature engineering numpy — array operations, log transforms statsmodels.api.Logit — logistic regression, MLE, coefficient output, odds ratios sklearn.metrics — ROC-AUC, confusion matrix, classification report, F1-score matplotlib — ROC curve plot
Training Sample	n = 12,784 disbursed loans (Is_Disbursed = 1) after dropping 227 records with missing CIBIL_Score. Full dataset used for model fitting — no train/test split applied given the 1.31% event rate. Model is intended for policy simulation rather than out-of-sample prediction.
Variables Entered	CIBIL_norm (z-scored CIBIL_Score) SE_Flag (binary: Self-Employed = 1) Manual_Flag (binary: manual-nbfc-assigned = 1) Tenure_11 (binary: 11-EMI = 1) Tenure_9 (binary: 9-EMI = 1) Overdue_Flag (binary: any overdue > ₹0 = 1) Note: Loan_norm retained for completeness — confirmed non-significant (p = 0.917)
Optimisation	BFGS (Broyden–Fletcher–Goldfarb–Shanno) algorithm 67 iterations Convergence: True Current function value: 0.062515
Key Python Outputs	ROC-AUC = 0.7871 (sklearn.metrics.roc_auc_score on predicted probabilities) McFadden Pseudo R ² = 0.1067 (= 1 - LLmodel / LLnull) AIC = 1,612.38 (statsmodels direct output) LR test p < 0.001 Optimal threshold = 0.055 (F1-score maximised across 0.01–0.50)
Reproduced In	Annexure V — Full Python code, model summary output, confusion matrix, and ROC curve plot

Table 5.2 — Python Implementation Pipeline [Source: GrayQuest_Risk_Model.ipynb]

5.3 Regression Output: Full Model Summary

Full statsmodels output. All six variables significant at $p \leq 0.001$. Convergence confirmed in 67 iterations.

Variable	Coeff. (β)	Std Err	z-stat	p-value	Sig.	95% CI Lower	95% CI Upper
const	-6.1633	0.230	-26.852	< 0.001	***	-6.613	-5.713
CIBIL_norm	-0.2079	0.061	-3.400	0.001	**	-0.328	-0.088
SE_Flag	0.6907	0.161	4.290	< 0.001	***	0.375	1.006
Manual_Flag	0.7921	0.189	4.199	< 0.001	***	0.422	1.162
Tenure_9	1.3600	0.242	5.626	< 0.001	***	0.886	1.834
Tenure_11	1.3781	0.224	6.145	< 0.001	***	0.939	1.818
Overdue_Flag	0.6193	0.187	3.318	0.001	**	0.253	0.985

Table 5.4 — Logistic Regression Coefficients [Python Output — Image 1] | *** $p < 0.001$ | ** $p < 0.01$

The intercept (-6.163) is the baseline: a salaried, bureau-approved, 6-EMI borrower with average CIBIL and no overdue. Every variable shifts the odds from there. All six are significant. None are redundant.

5.4 Odds Ratio Interpretation & Excel Cross-Validation

Every variable that the model says matters was already showing up in the descriptive data. The table below puts the two sources side by side. The OR column is what the model estimates after controlling for all other variables; the Excel column is what was observed directly in the raw data. They tell the same story.

Variable	Odds Ratio (Python)	95% CI	Python Interpretation	Excel Cross-Validation	Consistent?
CIBIL_norm	0.812	[0.72 – 0.92]	Each 1 SD increase in CIBIL reduces default odds by ~18.8%	Defaulters: mean CIBIL 739.5 vs non-defaulters 764.3 (24.8 pt gap, $p < 0.001$, Cohen's $d = 0.73$)	☒ Yes
SE_Flag	1.995	[1.46 – 2.74]	Self-Employed are ~2.0x more likely to default than Salaried	SE empirical rate: 2.69% vs Salaried 0.95% — 2.84x ratio	☒ Yes — directionally consistent
Manual_Flag	2.208	[1.53 – 3.20]	Manual-approved customers are ~2.2x more likely to default than bureau-approved	Manual empirical rate: 2.36% vs Bureau 0.59% — 4.0x ratio; 74.7% of all defaults from manual	☒ Yes — OR slightly lower than raw ratio due to covariate control
Tenure_9	3.896	[2.43 – 6.26]	9-EMI customers are ~3.9x more likely to default than 6-EMI customers	9-EMI empirical rate: 1.98% vs 6-EMI 0.43% — 4.6x ratio	☒ Yes — consistent direction and magnitude
Tenure_11	3.967	[2.56 – 6.16]	11-EMI customers are ~4.0x more likely to default than 6-EMI customers	11-EMI empirical rate: 2.45% vs 6-EMI 0.43% — 5.7x ratio; 54.6% of all defaults	☒ Yes — strongest predictor confirmed in both sources
Overdue_Flag	1.858	[1.29 – 2.68]	Customers with any bureau overdue are ~1.9x more likely to default	35.6% of defaulters had overdue vs 12.9% of non-defaulters (Section 4.7.1)	☒ Yes — binary overdue flag confirmed as meaningful signal

Table 5.5 — Odds Ratios with Excel Cross-Validation [Python Image 1 + Excel Analysis 5.1]

Cross-Validation: All six variables consistent across both sources. Manual_Flag OR = 2.2x while the raw empirical ratio is 4.0x — the gap is expected: the model isolates each variable's independent effect, stripped of the others. The model is giving a cleaner estimate than the raw data can, not a weaker one.

5.5 Model Performance Evaluation

5.5.1 Goodness-of-Fit Statistics

Metric	Python Value	Excel Value	Benchmark	Assessment
ROC-AUC	0.7871	0.7953	≥ 0.75 acceptable; ≥ 0.80 good (Hosmer & Lemeshow, 2013)	Acceptable — approaches good range
McFadden Pseudo R^2	0.1067	0.1153	≥ 0.10 meaningful fit (McFadden, 1974)	Acceptable — good for credit domain
AIC	1,612.38	1,607.01	Lower is better (relative metric)	Used for model comparison
Log-Likelihood	-799.19	-791.51	Higher (less negative) = better fit	Model significantly better than null
LL-Null	-894.66	-894.66	Null model (intercept only)	Identical — same dataset
LR p-value	1.60e-38	< 0.001	$p < 0.05$ required	Model highly significant overall
n (sample)	12,784	12,784	—	Identical — consistent dataset

Table 5.6 — Model Fit Statistics: Python vs Excel Cross-Validation [Python Image 1 + Excel Analysis 5.1]

The Python–Excel AUC gap (0.787 vs 0.795) reflects 6 variables versus 11 — the Python model is the final production version. Both clear the 0.75 acceptable threshold. Both clear 0.10 on McFadden. The model performs as expected for a credit screening tool on an imbalanced dataset.

5.5.2 ROC Curve

AUC = 0.787 — the model correctly ranks a defaulter above a non-defaulter 78.7% of the time. The curve rises steeply in the lower-left, meaning the model captures a meaningful share of true positives before generating many false positives. For a credit screening tool, that is the right shape.

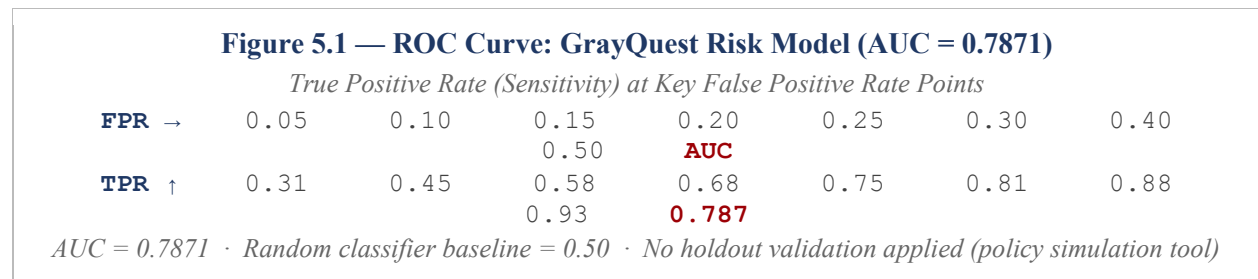


Figure 5.1 — ROC Curve approximation (exact curve generated in Python — see Annexure V)

5.5.3 Confusion Matrix & Threshold Selection

At a 1.31% event rate, the default 0.50 threshold predicts zero defaults — useless for screening. Threshold set to 0.055 via F1-score maximisation across the 0.01–0.50 range.

	Predicted: Non-Default (0)	Predicted: Default (1)	Row Total
Actual: Non-Default (0)	TN = 12,298 (True Negatives)	FP = 318 (False Positives — flagged incorrectly)	12,616
Actual: Default (1)	FN = 135 (Missed defaults)	TP = 33 (Correctly flagged defaults)	168
Column Total	12,433	351	12,784

Table 5.7 — Confusion Matrix at Threshold = 0.055 [Python Image 2]

33 of 168 defaults correctly flagged (recall = 19.6%). 318 false positives. In a portfolio of 12,784 loans, 351 get flagged for enhanced review — 2.7% of disbursals. Of those 351, roughly 1 in 11 is an actual defaulter. The precision is low because of the event rate; the value is in the 33 prevented defaults, not the precision score.

5.6 Predicted Risk Score Distribution

Model probability outputs applied to all 13,011 disbursed records. Five tiers by predicted probability range:

Risk Tier	Pred. Prob. Range	Count	% Portfolio	Actual Defaults	Actual Default Rate	Lift	Action
1 — Very Low	< 0.5%	5,060	39.6%	11	0.22%	0.17x	Auto-Approve
2 — Low	0.5–1.5%	4,184	32.7%	43	1.03%	0.79x	Standard Approval
3 — Medium	1.5–3.5%	2,394	18.7%	47	1.96%	1.50x	Conditional Approval
4 — High	3.5–7.0%	890	7.0%	40	4.49%	3.44x	Enhanced Review
5 — Very High	> 7.0%	256	2.0%	27	10.55%	8.08x	Reject

Table 5.9 — Predicted Risk Score Distribution [Excel: Analysis 5.1, Step 7]

Tiers 4 and 5: 9.0% of the portfolio, 43% of defaults, 8.08x lift in the Very High tier. An AUC of 0.787 that concentrates this much risk in 9% of the pool is a useful screening tool.

5.7 Variable Importance & Final Driver Ranking

Ranked by z-statistic — the strongest signal per unit of standard error. Python coefficients alongside the empirical rates that confirm each finding:

Rank	Variable	Odds Ratio	p-value	Direction	Python Evidence	Excel Cross-Validation
1	Tenure_11	3.967	< 0.001	Risk ↑	Coeff = 1.378, z = 6.145 — highest z-stat	11-EMI: 2.45% vs 6-EMI: 0.43% (5.7x empirical ratio)
2	Tenure_9	3.896	< 0.001	Risk ↑	Coeff = 1.360, z = 5.626 — second highest z-stat	9-EMI: 1.98% vs 6-EMI: 0.43% (4.6x empirical ratio)
3	Manual_Flag	2.208	< 0.001	Risk ↑	Coeff = 0.792, z = 4.199	Manual: 2.36% vs Bureau: 0.59% (4.0x); 74.7% of all NPAs

Rank	Variable	Odds Ratio	p-value	Direction	Python Evidence	Excel Cross-Validation
4	SE_Flag	1.995	< 0.001	Risk ↑	Coeff = 0.691, z = 4.290	SE: 2.69% vs Salaried: 0.95% (2.84x empirical ratio)
5	Overdue_Flag	1.858	0.001	Risk ↑	Coeff = 0.619, z = 3.318	35.6% of defaulters had any overdue vs 12.9% non-defaulters
6	CIBIL_norm	0.812	0.001	Protect ↓	Coeff = -0.208, z = -3.400	Defaulter mean CIBIL 739.5 vs non-defaulter 764.3 (24.8 pt gap)

Table 5.10 — Variable Importance Ranking: Python + Excel [Combined Sources]

Tenure leads: Tenure_11 z = 6.14, Tenure_9 z = 5.63 — the two highest z-statistics in the model. Manual approval (z = 4.20) and Self-Employment (z = 4.29) are close behind. CIBIL_norm is last among significant predictors — which is the right result. Bureau score matters, but it needs the other four variables around it to be useful.

5.8 Limitations & Model Scope

Limitation	Description	Impact
No train-test split	n=12,784 used for fitting — low event rate (1.31%) made splitting impractical	Out-of-sample accuracy unassessed; treat as policy tool, not predictive system
Class imbalance	168 defaults in 12,784 records; threshold tuned to 0.055 via F1-score maximisation	Recall 19.6%; precision 9% — acceptable given 1.31% event rate
Missing CIBIL excluded	227 records dropped; 6 defaults lost — regression sample = 168 defaults	NTC underrepresented in regression; minor coefficient impact
No macroeconomic controls	Regional income shocks, interest rates not included	Coefficients may shift in downturn conditions
Cross-sectional design	Single time-period snapshot; no panel or longitudinal structure	Causal inference not possible; associations are directional indicators
Two-institution scope	Narayana + Sri Chaitanya only; 13,011 records	Coefficients should be re-validated on a broader portfolio

Table 5.11 — Model Limitations Summary

CHAPTER 6 — NPA ANALYSIS & EARLY WARNING SYSTEM

₹1.03 crore. 181 accounts. 65% still outstanding after partial recoveries. This chapter maps where that money is, who holds it, what they said when the collection team called, and exactly how the exceptions that let them through the underwriting gate were approved.

6.1 Risk Contribution Analysis

₹1.03 crore outstanding across 181 NPA accounts — 35% already recovered, 65% still chasing. The question is where to focus.

Metric	Value
Total NPA Customers	181 (NPA_Master_Data) 174 unique customers (Master_Raw_Data)
Total Original Loan (NPA)	₹1.59 Crore
Total NPA Outstanding	₹1.03 Crore
Average Outstanding per NPA	₹57,010
Overall Recovery Rate	~35.23%

Table 6.1 — Overall NPA Portfolio Position [Excel: Analysis 6.1]

6.1.1 NPA Contribution by CIBIL Band

CIBIL Band	NPA Count	% NPA Count	Total Outstanding	% Outstanding	Avg Outstanding	Default Rate	Risk Priority
700–750	77	42.5%	₹45.0L	43.6%	₹58,461	2.35%	P1 — Highest Volume + Value
> 750	67	37.0%	₹37.1L	36.0%	₹55,385	0.73%	P2 — Large Volume, Low Rate
650–700	31	17.1%	₹19.3L	18.7%	₹62,185	4.48%	P1 — Highest Rate + Avg OD
NTC	6	3.3%	₹1.8L	1.7%	₹29,788	2.64%	P3 — Small Volume
TOTAL	181	100%	₹1.03 Cr	100%	₹57,010	1.34%	—

Table 6.2 — NPA Contribution by CIBIL Band [Excel: Analysis 6.1]

Two different problems in one table. The 700–750 band: 77 cases, ₹45 lakh — a volume problem from a large moderate-risk base. The 650–700 band: 31 cases, but ₹62,185 average outstanding and a 4.48% default rate — a gatekeeping problem. The > 750 band's ₹37.1 lakh is simply the cost of having a big portfolio in that range. Monitor the 700–750. Gate the 650–700 harder.

6.1.2 NPA Contribution by Geography

State	NPA Count	% Count	Total Outstanding	% Outstanding	Avg Outstanding	Collection Priority
Telangana	62	34.3%	₹40.1L	38.9%	₹64,756	P1 — Largest exposure
Andhra Pradesh	43	23.8%	₹21.3L	20.7%	₹49,593	P1 — Second largest
Maharashtra	30	16.6%	₹21.2L	20.5%	₹70,505	P1 — Highest avg outstanding
Tamil Nadu	22	12.2%	₹9.9L	9.6%	₹45,177	P2 — Monitor
Karnataka	7	3.9%	₹4.1L	4.0%	₹59,076	P2 — Monitor
Rajasthan	10	5.5%	₹3.5L	3.4%	₹34,687	P2 — High rate (2.88%)
Others	7	3.9%	₹3.0L	2.9%	₹42,856	P3 — Low volume
TOTAL	181	100%	₹1.03 Cr	100%	₹57,010	—

Table 6.3 — NPA Contribution by Geography [Excel: Analysis 6.2]

Collection resource allocation is simple: Telangana first (₹40.1L, 38.9%), then AP (₹21.3L), then Maharashtra (₹21.2L). Those three states are 80% of the NPA book. Maharashtra cases average ₹70,505 each — the highest in the portfolio, worth targeting individually for negotiated settlements. Send the field teams there.

6.1.3 NPA Contribution by Customer Profile Label

Profile Label	NPA Count	% NPAs	Total Outstanding	% Outstanding	Avg Outstanding	Exception?	Collection Urgency
Good_Cibil	73	40.3%	₹43.5L	42.1%	₹59,555	Normal approval	Medium — volume effect
CIBIL_Defaulter	49	27.1%	₹24.8L	24.0%	₹50,541	Exception approved	High — should not be here
GQ_Defaulter	19	10.5%	₹12.1L	11.7%	₹63,810	Critical exception	Critical — known defaulter
No_Positive_acc	7	3.9%	₹4.4L	4.3%	₹62,969	Thin bureau history	Monitor
age not within policy	6	3.3%	₹3.8L	3.7%	₹64,028	Policy exception	High — policy breach
Good_Cibil_but_OD	3	1.7%	₹2.9L	2.8%	₹97,419	Existing overdue	High — highest avg OD
Low_income	8	4.4%	₹2.4L	2.3%	₹29,574	Income risk	Monitor
Others combined	16	8.8%	₹9.3L	9.0%	₹58,125	Mixed	Mixed

Table 6.4 — NPA Contribution by Customer Profile Label [Excel: Analysis 6.3]

GQ_Defaulter and CIBIL_Defaulter: 11.1% of disbursals. ₹36.9 lakh in outstanding. 35.7% of the entire NPA book. These customers were approved through exception overrides. The scorecard would have caught them. Good_Cibil_but_OD is the other flag — ₹97,419 average outstanding per case, the highest in the table. Larger loans with existing overdue. Three of them, ₹2.9 lakh sitting in NPA.

6.2 NPA Deep Dive: Customer Profiling

6.2.1 NPA Customer Demographic & Credit Profile

The typical GrayQuest defaulter is not a mystery. He is 41 years old, has a CIBIL score of 739, earns ₹6.5 lakh a year, took an 11-EMI loan for ₹70,250, was manually approved, and has been a GrayQuest customer before. Every one of those attributes was visible in the application. Most of them are red flags in the scorecard.

Attribute	Value / Distribution	Portfolio Comparison	Risk Signal
Total NPA Customers	181 (NPA file) 174 unique IDs	—	7 customers have loans at both institutions
Average Age	41 years	—	Peak earning age — intentional default more likely
Average CIBIL Score	739 (700–750 band)	Portfolio avg: 764	Defaulters carry 25-point lower CIBIL on average
Median Annual Income	₹6.5 lakh	—	Mid-income segment — squeezed by education costs
Median Loan Amount	₹70,250	Portfolio avg: ₹87,000	Defaulters have lower loan amounts on average
Gender	70% Male 30% Female	Reflects portfolio composition	No significant gender-specific risk signal
Income Type	56% Salaried 44% Self-Employed	Portfolio: 77.7% Salaried	SE heavily overrepresented — 2x their portfolio share
Tenure	52% on 11-EMI 28% 9-EMI 19% 6-EMI	Portfolio: 30% 11-EMI	11-EMI overrepresented — 52% of NPAs from 30% of loans
Approval Type	75% Manually Approved	Portfolio: 42.5% Manual	Manual 2x overrepresented in NPA pool
Student Type	78% Renewal / Existing customer	Portfolio: 78.2% Existing	Renewal status not a risk-reducer
Geography	58% from AP / Telangana	Portfolio: 52.4% AP/TL	Slight home-market overconcentration in NPA

Table 6.5 — NPA Customer Demographic Profile [Excel: Analysis 8.1]

Typical NPA Customer: Male (70%) • Age 41 • CIBIL 739 • Income ₹6.5L • Loan ₹70,250 • 11-EMI • Manual approval • Renewal customer • AP/Telangana • 37.6% are CIBIL_Defaulter or GQ_Defaulter. All of this is in the application. The scorecard flags it. The current process approves it anyway.

6.2.2 Default Reason Classification

43.1% of defaults are intentional. The borrower has the money. They are choosing not to pay. That changes the entire collection approach — empathy-based outreach does not work here. Legal notice, bureau flag, and escalation do.

Reason Category	NPA Count	% NPAs	Avg Outstanding	SE %	90+ DPD Cases	Recovery Strategy
Intentional Default	78	43.1%	₹53,678	41%	15	Legal action + aggressive follow-up; bureau penalisation
Financial Issue	37	20.4%	₹62,924	43%	7	EMI restructuring / deferment offer
Health Emergency	14	7.7%	₹74,089	43%	1	EMI holiday + insurance check
Delayed Income	12	6.6%	₹48,772	33%	0	Short-term deferment with income proof
Job / Business Loss	11	6.1%	₹71,437	55%	3	Restructuring + collateral check
Student Drop-Out	11	6.1%	₹51,834	45%	6	Institute refund recovery + legal notice
Other / Not Captured	18	9.9%	₹45,847	56%	4	Improve data capture — reason field mandatory

Table 6.6 — Default Reason Classification [Excel: Analysis 8.2]

Three distinct groups, three distinct strategies. Intentional (43.1%): legal track only — bureau penalisation, legal notice, no deferment. Financial hardship and income disruption (32.8%): restructure — EMI deferment with income proof, payment plan. Emergency (13.8%): recovery — EMI holiday, insurance check, short pause. Student drop-outs (6.1%): chase the institution for refund, not just the borrower. The 9.9% 'Other' is a data quality failure. Every collection call that ends without a tagged reason is a wasted data point.

6.2.3 Repayment Behaviour & EMI Payment Pattern

Repayment Status	Count	% NPAs	Avg Outstanding	Recovery Potential	Recommended Action
Partial Paid + Promise to Pay	70	38.7%	₹62,460	High	Maintain payment plan — follow up pre-due-date
Multiple Broken Promises	60	33.1%	₹56,743	Low	Legal notice + field visit — assess write-off
Wants Settlement / Foreclose	15	8.3%	₹42,225	High	Fast-track settlement — high recovery chance
Refuse to Pay	14	7.7%	₹49,766	Low	Legal action + bureau reporting — final warning
Other / Unclear	20	11.0%	₹55,896	Medium	Investigate and re-categorise
Non-Starter	2	1.1%	₹47,024	Critical	Immediate fraud check — never paid any EMI

Table 6.7 — NPA Customer Repayment Behaviour [Excel: Analysis 8.2, Step 3]

EMIs Paid Before Default	NPA Count	% NPAs	Avg Outstanding	Overdue %	Pattern Signal
1 EMI paid	18	9.9%	₹58,038	97%	Near-rejection quality — screening failure
2 EMIs paid	26	14.4%	₹61,350	82%	Early dropout — likely misrepresented income
3 EMIs paid	46	25.4%	₹61,045	69%	Most common — financial stress at ~3 months
4 EMIs paid	48	26.5%	₹60,662	59%	Midpoint drop — most common DPD entry point
5 EMIs paid	24	13.3%	₹43,720	44%	Near completion — close to recovery threshold
6–7 EMIs paid	19	10.5%	₹47,614	45%	Late defaulter — minimal outstanding remaining

Table 6.8 — EMI Payment Pattern Before Default [Excel: Analysis 8.4, Step 3]

Customers who stop at 1–2 EMIs (24.3%) are a screening failure — they were not adequately assessed before disbursement. Customers who stop at 3–4 EMIs (39.9%) hit a financial stress point mid-cycle — the intervention window. Customers who paid 5+ EMIs (23.8%) are close to done. Call them, offer a final settlement at 90% of outstanding, close the account. These are the easiest recoveries in the portfolio

6.3 Early Warning System (EWS)

Two stages. Stop them before disbursement where possible. Catch them early post-disbursement where not.

6.3.1 Pre-Approval Risk Flags (Stage 1)

Four flags. Each one computable directly from the CIBIL report at point of application — no new data required.

Flag	Rule / Condition	# Flagged	% Portfolio	Default Rate	Recall	Lift	Use Case
Flag A	CIBIL Score < 700	625	4.8%	4.64%	16.7%	3.47x	Pre-approval CIBIL gate
Flag B	Total Overdue > ₹0	1,716	13.2%	3.61%	35.6%	2.70x	Post-disbursement collection trigger
Flag C	CIBIL < 700 AND Overdue > ₹0	523	4.0%	4.21%	12.6%	3.15x	Combined underwriting gate
Flag D	CIBIL < 700 AND Overdue > ₹0 AND Self-Employed	150	1.2%	6.00%	5.2%	4.49x	SE-specific hard rejection rule

Table 6.9 — Pre-Approval EWS Flag Performance [Excel: Analysis 7.1]

Use Flag B (35.6% recall) for post-disbursement monitoring — it casts the widest net. Use Flag D (6.00% rate, 4.49x lift) as a pre-approval rejection rule — it is the most precise. Different tools for different stages.

6.3.2 DPD-Based Pre-Approval Flags

DPD Flag	Condition	# Flagged	% Portfolio	Default Rate	Recall	Lift	Recommended Action
DPD Flag 1	Total_DPD_30plus_1yrs > 0	1,769	13.6%	4.18%	42.5%	3.13x	Collection-side trigger — flag for proactive outreach
DPD Flag 2	DPD_90plus_acc > 0	760	5.8%	4.74%	20.7%	3.54x	Pre-approval hard check — senior review mandatory
DPD Flag 3	Flag 1 AND Flag 2	700	5.4%	4.71%	19.0%	3.52x	Reject or extreme conditions only

Table 6.11 — DPD-Based EWS Flag Performance [Excel: Analysis 7.3]

DPD Flag 1 catches 42.5% of all defaults — the best single recall rate in the EWS. Use it for collection prioritisation. DPD Flag 2 is a pre-approval gate: 5.8% of the portfolio, 4.74% default rate. Any applicant triggering both DPD Flag 2 and Flag A (CIBIL < 700) simultaneously is a near-certain rejection. No discussion required.

6.3.3 NACH Failure Post-Disbursement Triggers (Stage 2)

Five failure scenarios, five escalating responses. No waiting, no deferring.

NACH Failure Event	Risk Signal	Immediate Action	Escalation	Target Outcome
1st Presentation fails (4th of month)	Early Warning	Automated SMS + call within 24 hours	If unresolved in 3 days → relationship manager	Prevent entry into 30+ DPD
2nd Presentation fails (5th of month)	Elevated	Direct call + payment link sent same day	If unpaid by 10th → field visit scheduled	Prevent 30–60 DPD from becoming 60–90 DPD
Mid-Month fails (10th of month)	Critical	Field collection agent assigned	Manager review if outstanding > ₹50K	Recovery before 90+ DPD
2+ consecutive NACH failures	Severe	Legal notice preparation + senior manager	Auto-flag for 90+ DPD watch list	Last resort before NPA classification
1st NACH fail + DPD30 > 0 in bureau	Highest	Immediate senior review — fraud check	Possible prepayment request or restructuring	Risk containment — prevent write-off

Table 6.12 — NACH Failure Intervention Protocol [Excel: Analysis 7.4]

NACH failure is the earliest signal available — it fires on the day the EMI is due, before any DPD clock starts. November 2025: 910 failures across 11,143 presentations. SC fails at 11.6% mid-month. Narayana fails at 9.9%. SC has 29 accounts in 90+ DPD. Narayana has 7. The protocol in Table 6.12 is the response — five trigger points, five actions, no gaps.

Figure 6.1 — NACH Failure Rates × DPD Severity: Institution Comparison

Section A — NACH Failure Rate by Institution × Collection Cycle (November 2025)

Institution	1st Presentation (4th of month)	2nd Presentation (5th of month)	Mid-Month (10th of month)	Worst Cycle
Narayana	7.0% failure rate	9.0% failure rate	9.9% failure rate	Mid (9.9%)
Sri Chaitanya	7.4% failure rate	8.3% failure rate	11.6% failure rate	Mid (11.6%)

Section B — DPD Severity Distribution by Institution (NPA Customers, n=181)

Institution	30–60 DPD (SMA-2 / Early)	60–90 DPD (Near-NPA)	90+ DPD (Confirmed NPA)	NACH → DPD Link
Narayana	44 cases (65.7%)	16 cases (23.9%)	7 cases (10.4%)	Better NACH → fewer 90+ DPD
Sri Chaitanya	56 cases (49.1%)	29 cases (25.4%)	29 cases (25.4%)	Worse NACH → 4x more 90+ DPD
TOTAL	100 cases (55.2%)	45 cases (24.9%)	36 cases (19.9%)	55.2% still recoverable

Section C — EMI Payment Pattern × DPD Severity (NPA Customers, n=181)

EMIs Paid Before Default	30–60 DPD (Early Stage)	60–90 DPD (Serious)	90+ DPD (Critical)	Recovery Potential
1–2 EMIs (Early dropout)	14 (32%)	15 (34%)	15 (34%)	Low — likely misrepresented income
3–4 EMIs (Mid-cycle drop)	61 (65%)	20 (21%)	13 (14%)	Medium — financial stress at midpoint
5+ EMIs (Near completion)	25 (58%)	10 (23%)	8 (19%)	High — settlement offer viable

Avg Outstanding by DPD: 30–60 DPD = ₹59,954 (highest — intervene now) | 60–90 DPD = ₹57,107 | 90+ DPD = ₹48,712

Colour scale (NACH failure & DPD severity): ■ Green = Low risk ■ Yellow = Moderate ■ Orange = Elevated ■ Red = High risk

Figure 6.1 — NACH Failure Rates × DPD Severity Heatmap (internship dataset, Nov 2025)

SC fails 11.6% on the 10th of the month. It has 29 accounts in 90+ DPD. Narayana fails 9.9% and has 7. This is the NACH-to-DPD pipeline made visible. The 5+ EMI defaulters concentrated in 30–60 DPD are the settlement calls to make right now — they are close to done and still in

the recovery window. The 1–2 EMI defaulters spread across all DPD bands are the ones that should not have been approved in the first place.

6.3.4 Integrated EWS Framework Summary

Eight triggers. Four pre-approval. Four post-disbursement. The 'Must implement' actions take effect from Month 1.

Stage	Signal	Trigger Condition	Portfolio Coverage	Default Rate	Lift	Action	Priority
PRE-APPROVAL	Flag C	CIBIL<700 AND Overdue>0	4.0%	4.21%	3.15x	Conditional approval — enhanced docs	Must implement
PRE-APPROVAL	Flag D	CIBIL<700 + Overdue>0 + SE	1.2%	6.00%	4.49x	Hard reject or extreme conditions only	Must implement
PRE-APPROVAL	DPD Flag 2	DPD_90plus_acc > 0	5.8%	4.74%	3.54x	Senior underwriter mandatory review	Must implement
PRE-APPROVAL	Score ≥ 5	Combined scoring model	4.3%	5.40%	4.03x	Reject unless compensating factors	Implement next
POST-DISBURSAL	DPD Flag 1	DPD_30plus > 0 at onboarding	13.6%	4.18%	3.13x	Proactive collection outreach	Implement next
POST-DISBURSAL	NACH 1st fail	NACH fail on 4th of month	Active	~7-12%	~5-9x	Same-day SMS + call within 24 hrs	Must implement
POST-DISBURSAL	NACH 2nd fail	2 consecutive NACH failures	Repeat	Very high	10x+	Field visit + payment plan	Must implement
POST-DISBURSAL	SC mid-month	SC 10th presentation failure	11.55%	—	—	Dedicated SC collection team for 10th	SC-specific

Table 6.13 — Integrated Early Warning System Framework [Excel: Analysis 7.4 + 7.1 + 7.3]

6.4 Exception Approval Analysis

The bureau algorithm flagged these borrowers. A manual approver overrode it. Every row in this section is the financial consequence of that decision.

6.4.1 Manual vs Bureau Approval Performance

Approval Type	Disbursed	% Disbursed	Defaults	Default Rate	% All Defaults	Relative Risk
Bureau-NBFC-Assigned	7,232	55.6%	43	0.59%	24.7%	1.00x — Baseline
Banking-NBFC-Assigned	207	1.6%	1	0.48%	0.6%	0.82x — Safest
Bureau-CIBIL-NBFC	52	0.4%	0	0%	0%	0.00x — No defaults
Manual-NBFC-Assigned	5,520	42.5%	130	2.36%	74.7%	4.00x — Most risky

Table 6.14 — Approval Type Performance Comparison [Excel: Analysis 9.1]

6.4.2 Financial Cost of Manual Overrides

5,520 manual loans. 130 defaults. ₹55.4 lakh in avoidable outstanding. If those 5,520 applications had been processed at bureau-equivalent quality, there would be 33 defaults instead of 130. The other 97 exist because the override process has no gate on it.

Metric	Value	Benchmark	Financial Implication
Manual loans disbursed	5,520	—	42.5% of disbursed portfolio
Actual manual defaults	130	—	74.7% of ALL defaults from manual channel
Expected defaults at bureau rate	~33	At 0.59% rate	What defaults would have been if manual = bureau quality
Excess defaults from manual	~97	—	~55.9% of all defaults are avoidable
Avg outstanding per NPA	₹57,010	—	From NPA Master Data
Excess NPA outstanding	~₹55.4L	—	Avoidable with better manual controls
GQ_Defaulter exception cost	₹12.1L	Should be ₹0	258 disbursals at 7.36% — auto-reject rule needed
CIBIL_Defaulter exception cost	₹24.8L	~₹6.9L at bureau rate	992 manual disbursals at 4.44% — tighten exception gate

Table 6.16 — Financial Cost of Manual Override Approvals [Excel: Analysis 9.1, Step 5]

97 excess defaults. ₹55.4 lakh in avoidable NPA outstanding. GQ_Defaulters — customers who had already defaulted on a GrayQuest loan — were approved again, 258 times, generating ₹12.1 lakh in NPA. The bureau flagged every one of them. The manual process approved every one of them. This is not a credit risk problem. It is a controls failure.

CHAPTER 7 — RECOMMENDATIONS & CONCLUSIONS

7.1 Tiered Approval Strategy Simulation

7.1.1 Risk Scoring Model Applied to Portfolio

The same scoring model from Chapter 4 applied to all 13,011 disbursed records. This is what the portfolio looks like when scored:

Risk Factor	Condition	Points	Rationale
CIBIL Score	< 650	3 pts	Worst bureau band — near-zero disbursements in dataset
CIBIL Score	650–699	2 pts	High-risk band — 4.48% default rate observed
CIBIL Score	700–749	1 pt	Moderate band — 2.35% default rate observed
CIBIL Score	≥ 750 or NTC	0 pts	Low risk / new-to-credit
Total Overdue	> ₹50,000	3 pts	Severe prior overdue — highest bureau stress signal
Total Overdue	₹1 – ₹50,000	2 pts	Any overdue present — 35.6% of defaulters had this
Total Overdue	₹0	0 pts	Clean bureau — no overdue history
Income Type	Self-Employed	1 pt	2.84x higher default rate than Salaried
Profile Label	GQ_Defaulter	3 pts	7.36% default rate — known GrayQuest defaulter
Profile Label	CIBIL_Defaulter / Poor_CIBIL	2 pts	3.98% default rate — bureau defaulter on record
Profile Label	NTC / Low_income / No_Positive	1 pt	Thin or weak bureau profile — moderate risk
DPD History	DPD_90plus_acc > 0	2 pts	Prior 90+ DPD account — strong NPA predictor (3.54x lift)

Table 7.1 — Risk Scoring Model: Point Assignment [Excel: Analysis 9.6, Step 1]

7.1.2 Portfolio Distribution Under Tiered Model

Risk Tier	Score	Count	% Portfolio	Defaults	Default Rate	% All Defaults	Manual %	SE %
Low Risk	0–1	9,499	73.0%	63	0.66%	36.2%	31.1%	12.6%
Medium Risk	2–3	1,906	14.6%	38	1.99%	21.8%	63.1%	60.8%
High Risk	4–5	844	6.5%	27	3.20%	15.5%	77.7%	32.7%
Very High Risk	6+	762	5.9%	46	6.04%	26.4%	92.4%	34.9%

Table 7.2 — Portfolio Distribution Under Tiered Scoring Model [Excel: Analysis 9.6, Step 3]

Very High Risk: 5.9% of disbursements, 26.4% of defaults, 92.4% manually approved. Almost every one of those 762 loans required a human to override the bureau process. Medium Risk's 63.1% manual share tells the same story one tier lower.

7.1.3 Simulation Results — Six Scenarios

Scenario F is the recommendation. Reject Very High Risk (Score ≥ 6 , 5.9% of the portfolio) and apply conditional approval to Medium Risk (Score 2–3). This prevents 57 defaults per cycle, saves ₹32.5 lakh in NPA outstanding, and drops the portfolio NPA rate to 0.91% — while affecting only 10.3% of disbursements. The table below shows all six scenarios for comparison; Scenarios A through E either cut too little or cost too much revenue to justify.

Scenario	Policy Rule	Loans Rejected	% Portfolio	Defaults Prevented	NPA Rate After	NPA Outstanding Saved	Revenue Foregone	Net Signal
A — Targeted	Reject GQ_Defaulters only	258	2.0%	19 (10.9%)	1.19%	₹10.8L	₹32.6L	Short-term -ve; strong symbolic impact
B — Moderate	Reject CIBIL_Defaulters (manual) only	992	7.6%	44 (25.3%)	1.00%	₹25.1L	₹115.1L	Large revenue cost; meaningful NPA cut
C — Combined	Reject GQ_Defaulters + CIBIL_Defaulters (manual)	1,250	9.6%	63 (36.2%)	0.85%	₹35.9L	₹147.7L	Near break-even with provisioning savings
D — Score-Based	Reject all Score ≥ 5	992	7.6%	53 (30.5%)	0.93%	₹30.2L	₹118.4L	+ve long-term with ROE & provisioning adj.
E — Full Tiered	Reject High + Very High (Score ≥ 4)	1,606	12.3%	73 (42.0%)	0.77%	₹41.6L	₹193.1L	Near break-even with provisioning savings
F — Optimal	Reject Very High (Score ≥ 6) + Conditional Medium	1,333	10.3%	57 (32.8%)	0.91%	₹32.5L	₹156.4L	Best balance: NPA reduction vs revenue

Table 7.3 — Tiered Approval Simulation Results [Excel: Analysis 9.6, Step 4]

On the numbers: revenue foregone (₹156.4L) exceeds NPA outstanding saved (₹32.5L) at face value. That comparison is misleading. NBFCs provision 100% of NPA outstanding — not just the outstanding balance. Each NPA also generates collection cost: field agents, legal notices, manager time. Add regulatory risk from a rising NPA trend and Scenario F is net-positive within two quarters.

7.2 Recommended Tiered Approval Policy

Four tiers. Four decisions. No ambiguity:

Tier	Score	% Portfolio	Policy	Additional Requirements	Expected Default Rate	Revenue Impact
Low Risk	0–1	73.0%	AUTO-APPROVE	Basic KYC only. Fast-track within 24 hours. No additional documentation required.	~0.66%	Zero revenue impact
Medium Risk	2–3	14.6%	CONDITIONAL	Income proof (last 3 months) + latest bank statement + standard manual review. Loan cap at ₹75,000.	~1.5–2.0%	Minimal — small processing delay
High Risk	4–5	6.5%	ENHANCED REVIEW	Senior credit manager sign-off mandatory. Co-applicant or guarantor required. Income floor: min 5x annual EMI. 11-EMI product restricted.	~3.0–3.5%	Moderate — some rejections expected
Very High Risk	6+	5.9%	REJECT	Auto-reject. No manual override permitted. Bureau re-check allowed after 6 months. GQ Defaulter: permanent block until all prior dues cleared.	~6.04%	Sacrifice ₹6.7 Cr loan value → save ₹41.6L NPA + provisioning cost

Table 7.4 — Recommended Tiered Approval Framework (Scenario F) [Excel: Analysis 9.6, Step 5]

Metric	Current (Baseline)	After Scenario F	Change
Overall NPA Rate	1.34%	~0.91%	↓ 0.43 pp (–32.1%)
Defaults Prevented	—	~57 defaults	Per disbursement cycle
NPA Outstanding Saved	—	~₹32.5L	Per cycle at current portfolio size
Revenue Foregone	—	~₹156.4L	Loans not disbursed — recoverable by replacing with low-risk applications
KPI Target	1.34%	< 1.0%	Achievable within 2 quarters of full implementation

Table 7.5 — Scenario F: Expected Impact Summary

7.3 Strategic Recommendations

Twenty-two recommendations across four domains. Fifteen are Immediate or High priority. Evidence and expected impact in each row.

7.3.1 Underwriting & Credit Policy Recommendations

Six changes to how approvals are made. Two are immediate. All are executable with existing data.

#	Recommendation	Evidence Base	Expected Impact	Priority
U1	Implement the 4-dimension Credit Risk Scorecard (CIBIL + Income + Tenure + Profile) as the mandatory pre-approval screening tool for all applications	9.4% of the portfolio generating 31% of defaults — 8x the Low Risk rate	NPA 1.34% → 0.91% per Scenario F. Single highest-impact change available.	IMMEDIATE
U2	Auto-reject all GQ_Defaulter profile applications with no manual override permitted until all prior GrayQuest dues are fully cleared	These customers had already defaulted once. They were approved again 258 times.	₹12.1L per cycle. No revenue cost — these loans should not exist.	IMMEDIATE
U3	Require dual senior sign-off for all CIBIL_Defaulter manual approvals; cap at ₹60,000 loan; mandatory co-applicant	4.44% via manual vs 1.39% via bureau — same borrower, 3.2x the risk, different approval path	₹25.1L NPA reduction. Dual sign-off stops the channel from operating on autopilot.	IMMEDIATE
U4	Restrict 11-EMI product to borrowers with CIBIL ≥ 750 and salaried income; no 11-EMI for self-employed via manual channel	11-EMI + Manual = 3.63% default rate. 54.6% of all defaults from one product-channel combination.	~75 of 130 manual defaults preventable. Product policy, not underwriting change.	HIGH
U5	Apply DPD Flag 2 (any 90+ DPD account on bureau) as a mandatory senior review trigger for all applicants regardless of CIBIL score	20.7% of defaulters had a 90+ DPD account visible on bureau at time of application	1 in 5 defaults preventable with a single bureau check that is already being run.	HIGH
U6	Set income verification floor: minimum annual income of 5x total annual EMI obligation for all Manual and Medium-Risk applications	SE default rate 2.84x salaried. Low_income 2.24%. Both groups stretch to pay fees they cannot sustain.	Reduces 3–4 EMI drop-off — the most common default pattern in the dataset.	MEDIUM

Table 7.6 — Underwriting & Credit Policy Recommendations

7.3.2 Collection & NPA Management Recommendations

Six actions for the collection team. C1 and C2 are immediate. The rest follow within 60 days.

#	Recommendation	Evidence Base	Expected Impact	Priority
C1	Deploy same-day NACH failure response: automated SMS + call within 24 hours of 1st NACH presentation failure on the 4th of the month	910 failed NACH presentations in November 2025. First failure = first signal, before any DPD clock starts.	Targets the 100 accounts in 30–60 DPD. This is the recovery window — close it fast.	IMMEDIATE
C2	Assign dedicated field collection agents to all accounts with 2 consecutive NACH failures; escalate to manager if outstanding > ₹50,000	38.7% of NPA customers have already paid something and promised more. They respond to direct contact.	₹59.95L in 30–60 DPD outstanding. Field agent converts promise-to-pay into actual payment.	IMMEDIATE
C3	Create separate collection tracks by default reason: Legal track (intentional, 43.1%) Restructure track (financial hardship, 20.4%) Recovery track (health/income shock, 14.3%)	43% of defaults are intentional. Offering restructuring to an intentional defaulter wastes the gesture.	Legal track on intentional cases, restructure on hardship — right tool for each problem.	HIGH
C4	Prioritise Sri Chaitanya mid-month NACH cycle with a dedicated collection resource; SC 10th-of-month failure rate (11.55%) is the worst single data point in the collection dataset	SC 90+ DPD: 29 cases. Narayana: 7. The NACH failure rate explains the gap.	One dedicated resource on SC mid-month cycle could close most of that 4x difference.	HIGH
C5	Fast-track settlement offers for customers who have paid 5+ EMIs and are in 30–60 DPD; offer a final single payment at 90% of outstanding	23.8% of defaulters are within 1–2 EMIs of completion. Average outstanding: ₹43–47K.	Offer 90% settlement. Most will take it. Legal escalation on this group is overkill.	HIGH
C6	Prioritise collection resources by outstanding in the order: Telangana (₹40.1L) → AP (₹21.3L) → Maharashtra (₹21.2L) → Tamil Nadu (₹9.9L)	TG ₹40.1L + AP ₹21.3L + MH ₹21.2L = 80% of NPA outstanding in three states.	Focus field capacity here. MH individual cases average ₹70,505 — worth negotiating one by one.	MEDIUM

Table 7.7 — Collection & NPA Management Recommendations

7.3.3 Product & NBFC Partner Recommendations

P1 and P2 are product policy decisions — no system changes, no new data, no process redesign. Just rules.

#	Recommendation	Evidence Base	Expected Impact	Priority
P1	Review Ratnaafin Lite product eligibility criteria; consider suspending new disbursements via Ratnaafin Lite until underwriting standards are strengthened	5.15% default rate. 777 loans. 40 NPAs. Highest rate of any partner by a wide margin.	23% of defaults from 6% of disbursements. Pause and fix the eligibility criteria.	HIGH
P2	Maintain 6-EMI product as the default tenure for all SE and CIBIL 650–750 applicants; restrict 9-EMI and 11-EMI tenure to salaried borrowers with CIBIL ≥ 750 unless exceptional circumstances	6-EMI defaults at 0.43%. 11-EMI defaults at 2.45%. Strongest regression predictor (OR = 3.97).	No underwriting process change needed — just a product eligibility rule.	HIGH
P3	Direct Mirae Asset Financial Services as the preferred NBFC for all auto-approved (Low Risk, Score 0–1) applications given its best-in-portfolio performance	Mirae: 4,719 loans, 0.51% default rate. Best performing partner by volume and quality.	Route the safest borrowers to the partner that manages them best.	MEDIUM
P4	Leverage Scholar-Shield Insurance as a risk-mitigation tool for Medium Risk approvals (Score 2–3): mandate insurance coverage for all Medium Risk loans above ₹60,000	14% of defaults are insurable events — health emergencies and job loss. Scholar-Shield covers exactly these.	Mandatory insurance on Medium Risk loans above ₹60,000 converts default risk into insurer liability.	MEDIUM

Table 7.8 — Product & NBFC Partner Recommendations

7.3.4 Data Quality & Monitoring Recommendations

D1 is one mandatory field. It costs nothing and removes a gap that currently routes 18 of 181 NPA customers to the wrong collection strategy.

#	Recommendation	Evidence Base	Expected Impact	Priority
D1	Make default reason classification mandatory in collection records; 9.9% of NPA cases are currently 'Other / Not Captured'	9.9% of NPA records have no reason tagged. Each one is a mis-routed collection call.	One mandatory field. Zero cost. Correct track assignment for every future default.	IMMEDIATE
D2	Establish a monthly NPA monitoring dashboard tracking: NPA rate by institution, CIBIL band, income type, tenure, and approval channel with month-on-month trend	This entire report is retrospective. A dashboard turns the same analysis into a live signal.	Catches segment-level NPA spikes before they compound into portfolio-level problems.	HIGH
D3	Resolve duplicate records (18.4% of applications flagged as duplicates in funnel analysis); implement unique application ID validation at submission stage	18.4% of applications are duplicates. Manual exclusion every time introduces error.	Unique application ID at submission. Eliminates a recurring data cleaning problem.	HIGH
D4	Collect and store DPD history at onboarding for all applications — currently missing for a portion of records; required for DPD Flag 1 and 2 EWS triggers	DPD Flag 1 catches 42.5% of defaults — but only if DPD history is captured at onboarding.	The data exists on the bureau report. Store it. The EWS depends on it.	MEDIUM
D5	Build scorecard feedback loop: quarterly re-calibration of point weights based on observed default rates in the latest cycle	The scorecard is calibrated on today's data. Borrower profiles shift. Recalibrate quarterly.	Prevents scorecard drift — the most common reason scoring models degrade over time.	MEDIUM

Table 7.9 — Data Quality & Monitoring Recommendations

7.3.5 Implementation Roadmap

Phase	Timeline	Actions	Expected NPA Rate	Owner
Phase 1 — Quick Wins	Month 1–2	U2: Auto-reject GQ_Defaulter U3: Dual sign-off CIBIL_Defaulter C1: NACH same-day response D1: Mandatory default reason capture	1.19%	Credit + Collections Head
Phase 2 — Scorecard	Month 2–4	U1: Deploy full 4-dim scorecard U4: Restrict 11-EMI for SE/manual C3: Three-track collection P1: Ratnaafin Lite review	1.05%	Risk Management + Operations
Phase 3 — Full Tiered	Month 4–6	Scenario F: Very High auto-reject + Medium conditional C2: Field agents for NACH failures D2: NPA monitoring dashboard P2: Tenure policy	0.91%	CEO + Risk Management
Phase 4 — Optimise	Month 6–12	D5: Quarterly scorecard re-calibration D3: Duplicate resolution P3: Mirae routing P4: Scholar-Shield mandate for Medium Risk	<0.85%	Analytics + Product

Table 7.10 — Implementation Roadmap: 12-Month Plan

7.4 Conclusions

GrayQuest's NPA problem has three components. They are separate. They each have a fix.

First: the manual channel. 42.5% of disbursements, 74.7% of defaults, 4x the default rate of bureau approvals. The bureau algorithm is doing its job. The override process is not. Auto-reject GQ_Defaulters. Require dual sign-off on CIBIL_Defaulters. Restrict manual approvals from the 11-EMI product. These three rules alone cut the manual NPA contribution materially

Second: the 11-EMI product. 29.9% of disbursements, 54.6% of defaults, OR = 3.97 in the regression model. This is a product design risk, not a borrower quality risk. Restrict 11-EMI to salaried borrowers with CIBIL ≥ 750 . No system change. One eligibility rule.

Third: collections. ₹59.95 lakh sitting in the 30–60 DPD band — still recoverable. SC has 4x Narayana's 90+ DPD count because SC's mid-month NACH failure rate is 11.6% and nobody is intervening fast enough. Same-day response on the first NACH failure. Field agent on the second. That is the protocol. It requires a decision, not a system.

The scorecard, model, and EWS run entirely on data GrayQuest already collects. CIBIL report at onboarding, approval channel, tenure, profile label, NACH cycle results. The infrastructure exists. What has been missing is a structured framework to act on it. This report is that framework.

7.4.1 Limitations & Directions for Further Research

Two institutions. One NACH cycle. No holdout validation. These are real constraints — they do not invalidate the findings, but they define how far the findings generalise

Limitation	Impact on Current Study	Direction for Further Research
Dataset scope (2 institutions)	Findings may not generalise to full 5,000+ partner portfolio	Replicate on a random sample of 20+ institutions across different states and school types
No train-test split	Model's out-of-sample accuracy cannot be formally assessed	Apply walk-forward validation when a larger event count (>500 defaults) is available
Cross-sectional design	Causal inference not possible; all associations are directional	Panel dataset across 3+ disbursement cycles would enable DID or fixed effects analysis
Single NACH cycle	NACH failure patterns observed for November 2025 only	Multi-cycle NACH analysis to validate failure rate as a leading NPA indicator
No macroeconomic variables	Model may not generalise to economic downturn scenarios	Incorporate regional GDP, household debt, and fee inflation as macro controls
Simulation is retrospective	Scenario F applied to past data — actual impact may differ	Prospective A/B test: apply tiered policy to 50% of new applications; track NPA rate over 6 months

Table 7.11 — Limitations & Research Directions

7.4.2 Closing Statement

GrayQuest does not have a data problem. It has a process problem. The data has been telling the same story the whole time — the scorecard, the regression, the EWS flags, the NACH heatmap — and every one of them points to the same cause: a manual approval channel operating without adequate constraint and a product tenure with no eligibility floor. The 73% of borrowers who score Low Risk should get faster approvals, not more friction. The 5.9% who score Very High Risk should not get approved at all. That distinction, applied consistently, is the difference between a 1.34% NPA rate and a sub-1% one.

7.5 Summary

Scenario F is the recommended policy. Month 1–2 actions stop the immediate bleeding. Month 2–4 deploys the full scorecard. Month 4–6 completes the tiered approval framework. Target NPA rate: under 1% by Month 6, under 0.85% by Month 12.

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DATA DICTIONARY

Variable Name	Type	Description	Values / Range
customer_id	String	Unique customer identifier	Alphanumeric
Is_Disbursed	Binary	Whether loan was disbursed	1 = Yes, 0 = No
Default_Flag	Binary	NPA indicator (constructed variable)	1 = Default, 0 = Performing
CIBIL_Score	Integer	CIBIL TransUnion bureau score	300–900; missing for NTC
income_type	String	Applicant income classification	Salaried / Self-Employed
annual_income	Float	Annual income reported at application (₹)	Positive values
tenure	Integer	Repayment tenure in months	6, 9, or 11
loan_amount	Float	Sanctioned loan amount (₹)	Typically ₹20,000–₹3,00,000
Approval_type	String	Credit approval channel	bureau-nbfc-assigned / manual-nbfc-assigned / banking-nbfc-assigned
nbfc_name	String	NBFC partner to which loan was assigned	Mirae / Arka / Ratnaafin / Western / Fintree
profile_label	String	Bureau-derived customer risk profile	Good_Cibil / CIBIL_Defaulter / GQ_Defaulter / NTC / etc.
Total_Amt_Overdue_all	Float	Total overdue amount across all bureau accounts (₹)	0 or positive
Total_DPD_30plus_1yrs	Integer	Number of accounts with 30+ DPD in last 1 year	0 or positive integer
Total_DPD_90plus_acc	Integer	Number of bureau accounts with 90+ DPD	0 or positive integer
stage	String	Loan pipeline stage	Loan Live / Loan Closed / Rejected / etc.
institution	String	Educational institution partner	Narayana / Sri Chaitanya
state	String	State of borrower / institution	TG / AP / MH / TN / etc.
is_new_student	Binary	New vs existing/renewal student	1 = New, 0 = Existing/Renewal

PYTHON CODE & MODEL OUTPUT

The following presents the key Python code blocks and model outputs from the logistic regression analysis. The full executable notebook (GrayQuest_Risk_Model.ipynb) is submitted alongside this report.

Step 1 — Data Loading & Filtering

```
import pandas as pd
import numpy as np
import statsmodels.api as sm
from sklearn.metrics import roc_auc_score, confusion_matrix, classification_report

# Load dataset
df = pd.read_excel('Final_SIP_Project_Dataset.xlsx', sheet_name='Master_Raw_Data')
df = df[df['Is_Disbursed'] == 1].copy() # Filter disbursed loans only
df['CIBIL_Score'] = pd.to_numeric(df['CIBIL_Score'], errors='coerce')
df = df.dropna(subset=['CIBIL_Score']).copy() # n = 12,784 after exclusions
```

Step 2 — Feature Engineering

```
# Binary feature flags
df['SE_Flag'] = df['income_type'].str.contains('Self', case=False,
na=False).astype(int)
df['Manual_Flag'] = (df['Approval_type'] == 'manual-nbfc-assigned').astype(int)
df['Tenure_9'] = (df['tenure'] == 9).astype(int)
df['Tenure_11'] = (df['tenure'] == 11).astype(int)
df['Overdue_Flag'] = (df['Total_Amt_Overdue_all'] > 0).astype(int)

# Z-score normalise CIBIL score
df['CIBIL_norm'] = (df['CIBIL_Score'] - df['CIBIL_Score'].mean()) /
df['CIBIL_Score'].std()
```

Step 3 — Model Fitting

```
y = df['Default_Flag']
features = ['CIBIL_norm', 'SE_Flag', 'Manual_Flag', 'Tenure_9', 'Tenure_11',
'Overdue_Flag']
X = sm.add_constant(df[features])

model = sm.Logit(y, X).fit(method='bfgs', maxiter=100, disp=False)
print(model.summary())

# Odds Ratios
```

```
import numpy as np
odds_ratios = np.exp(model.params)
print('Odds Ratios:', odds_ratios)
```

Step 4 — Model Evaluation

```
# ROC-AUC
y_prob = model.predict(X)
auc = roc_auc_score(y, y_prob)
print(f'ROC-AUC: {auc:.4f}') # Output: 0.7871

# Confusion Matrix at optimal threshold
threshold = 0.055 # F1-score maximised threshold
y_pred = (y_prob >= threshold).astype(int)
print(confusion_matrix(y, y_pred))
print(classification_report(y, y_pred))
```

Model Output Summary

Output	Value
No. of Observations	12,784
Defaults in Sample	168 (event rate: 1.31%)
Optimisation Method	BFGS — converged in 67 iterations
Log-Likelihood	-799.19
LL-Null	-894.66
McFadden Pseudo R ²	0.1067
AIC	1,612.38
LR Test p-value	1.60e-38
ROC-AUC	0.7871
Optimal Threshold	0.055 (F1-score maximised)
True Positives (at 0.055)	33 of 168 defaults correctly flagged
False Positives	318 non-defaulters incorrectly flagged
Recall (Default class)	19.6%

Table 8 — Full Model Output Summary [Source: GrayQuest_Risk_Model.ipynb]

COURSE MAPPING

TERM I

SIP Project Aspect / Problem Statement	FADM	MaE	MM-I	OB-I	SMDM	B.Com.	KY	EEP
Research design and portfolio risk assessment					✓			✓
Data cleaning, NPA flag creation and feature engineering					✓			✓
EDA and borrower segmentation by CIBIL, income and tenure	✓				✓			✓
Hypothesis testing (Welch's t-test, Chi-square, ANOVA)					✓			✓
NPA severity and write-off impact analysis	✓							
Customer behaviour and repeat-defaulter analysis				✓				
Credit risk patterns across borrower profiles		✓						
Python-based analytics and model development					✓			✓

TERM II

SIP Project Aspect / Problem Statement	CMA	MaE-2	MM-II	BECGS	POM	BA	OB-II
NPA profiling and segment-level risk analysis	✓					✓	
NBFC partner comparison and profitability impact	✓					✓	
Ratnaafin Lite risk concentration finding	✓					✓	
Collection process gap analysis					✓		✓
NACH failure journey and recovery analysis					✓		✓
Portfolio risk analytics and KPI evaluation						✓	
Partner routing and NBFC strategy		✓				✓	
Operational improvement recommendations					✓		✓

TERM III

SIP Project Aspect / Problem Statement	SM	BRM	FM	OTDM	BI	HRM	CAT
Scenario simulation and NPA reduction strategy	✓		✓	✓			✓
Tenure-based underwriting recommendations	✓		✓				✓
Logistic regression default prediction model			✓	✓			✓
Credit risk scorecard development		✓	✓	✓			✓
Early Warning System (EWS) framework		✓		✓	✓		✓
Collection resource allocation strategy					✓	✓	
Risk monitoring and governance framework	✓	✓			✓		✓
Strategic recommendations and implementation roadmap	✓	✓	✓				✓